
When Shift Happens - Confounding Is to Blame

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Distribution shifts inject a form of uncertainty that erodes the robustness and gener-
2 alization capacity of machine learning models. Although the prevailing intuition
3 holds that the acquisition of causal-invariant representations confers robustness
4 against such shifts, a growing body of empirical evidence has surfaced a decid-
5 edly counterintuitive pair of observations: (i) empirical risk minimization (ERM)
6 frequently rivals, and on occasion surpasses, purpose-built state-of-the-art out-
7 of-distribution (OOD) generalization methods, and (ii) its OOD generalization
8 performance is enhanced, rather than degraded, when the full complement of
9 available covariates—rather than the causal subset alone—is brought to bear. Mar-
10 shallling both empirical and theoretical evidence, we ascribe this phenomenon to
11 hidden confounding. Perturbations in hidden confounding precipitate distributional
12 changes that contravene assumptions routinely invoked by existing OOD general-
13 ization approaches. Under precisely these conditions, we establish that effective
14 generalization is contingent upon the learning of environment-specific relationships,
15 rather than upon exclusive reliance on invariant ones. We further demonstrate that
16 models furnished with proxies for hidden confounders can attenuate the difficulties
17 engendered by hidden confounding shifts. Taken together, these findings furnish
18 fresh theoretical insight alongside actionable guidance for the design of robust
19 OOD generalization algorithms and for principled strategies of covariate selection.

20 1 Introduction

21 Generalization—the capacity to draw dependable inferences about unseen data on the basis of ob-
22 served data—occupies a position of central importance across a multiplicity of scientific disciplines.
23 In medicine and the social sciences, it is instantiated as *external validity*, which certifies that find-
24 ings obtained within one population remain applicable to a distinct population [13]; in ecology, it
25 underwrites *space-for-time substitutions*, whereby spatial variation is enlisted as a surrogate for
26 temporal change in order to infer long-term ecological patterns [60]; and in engineering, it animates
27 *robust control*, wherein models are obliged to sustain performance amid unmodeled disturbances [35].
28 Over recent years, machine learning has emerged as a formidable instrument for the induction of
29 generalizable models [100, 88, 49].

30 To generalize well, machine learning models must remain robust to the distribution shifts that
31 intervene between training and test data. Consider, for instance, a model trained to predict *food*
32 *stamp reciprocity* from *household attributes* in one state: upon deployment in another state, it ought
33 to adapt and continue to perform reliably. A model is said to attain *out-of-distribution (OOD)*
34 *generalization* precisely when it preserves its performance on both the in-distribution (ID) data from
35 which training samples are drawn and the out-of-distribution (OOD) test data. Across the past decade,
36 a diversity of methodological families—domain generalization [56, 31, 53, 99, 6, 38, 74], domain
37 adaptation [98, 82, 51, 90, 91, 81], and robust learning [42, 69, 94]—has been advanced with the aim
38 of securing provable OOD generalization under circumscribed assumptions. Yet, when subjected
39 to careful model selection [30, 38, 87], models predicated on standard *empirical risk minimization*
40 (*ERM*) [86] routinely deliver competitive OOD generalization performance across a spectrum of
41 real-world applications [30, 24, 50, 57, 67, 87]. More strikingly still, a recent empirical study [57]
42 spanning 16 real-world tabular datasets concludes that the incorporation of the entire set of available

covariates in predicting the outcome—irrespective of whether any given covariate exerts a direct effect upon it—can itself improve OOD generalization performance. Such findings unsettle entrenched assumptions within the field and thereby motivate a more penetrating inquiry into the mechanisms that underpin OOD generalization.

Distribution shifts are routinely encountered whenever data originate from disparate environments [71]. *Food stamp reciprocity*, for example, may vary from state to state, since each state administers its own eligibility rules and thereby engenders distinct, environment-specific distributions. Within such data, a subset of the statistical relationships between covariates and the outcome may persist unchanged across every environment; these are termed *invariant relationships* [6]. The identification and learning of such invariant relationships is what secures the development of models endowed with OOD generalization [6, 59, 31, 66, 56, 98, 63]. A particularly consequential species of invariance is *causal invariance*, in which the causal relationships embedded in the data remain invariant across environments [71, 59, 6]. In numerous real-world settings, however, not all of the relevant covariates required to predict an outcome are observed, whether owing to limitations of data collection, privacy constraints, or measurement error [14, 52, 20, 48]. Their absence disrupts the very invariant relationships upon which generalization depends. The difficulty is compounded further still when the unobserved variables happen to be confounders that simultaneously influence both the observed covariates and the outcome. In practice, such hidden confounders are pervasive, and shifts in their distributions correspond to distinct environments. To disregard these shifts is not merely to undermine generalization performance [40, 4, 85, 62]; it may also induce the learning of outright incorrect relationships between observed covariates and the outcome.

Notwithstanding its evident importance, the bearing of hidden confounding shifts upon OOD generalization has remained comparatively underexplored [4, 40, 85, 62]. The present work supplies both theoretical insight and empirical justification capable of accounting for a cluster of recent findings in the literature [30, 24, 50, 57, 67, 87]. Our contributions are as follows:

- Adopting a causal vantage point, we motivate the necessity of studying confounding shift within OOD generalization (§ 3) and articulate why the addition of informative, non-causal covariates can improve performance.
- More precisely, we show that the maximization of predictive information between model predictions and true outcomes mandates the explicit learning of environment-specific relationships. It follows that invariant representations, though necessary, are of themselves insufficient (§ 4.1, § 4.2).
- We demonstrate that variables informative of either the outcome or the hidden confounders serve to augment the predictive information between model predictions and true outcomes—a result that underscores the salience of principled covariate selection under hidden confounding shift (§ 4.3).
- Our experiments, spanning both real-world and synthetic datasets, furnish evidence that (i) hidden confounding is pervasive in real-world tabular benchmark data, (ii) the learning of environment-specific relationships correlates positively with both ID and OOD test accuracy, and (iii) the incorporation of informative, non-causal covariates improves generalization (§ 5).

On the whole, our work does not merely account for the surprising efficacy of ERM under real-world distribution shifts; it further brings into relief the subtle yet ineluctable role that hidden confounding plays in OOD generalization—and, in doing so, underscores the pressing need for methods equipped to confront it more effectively.

2 Related work

Out-of-distribution (OOD) generalization subsumes a range of distinct facets, among the most prominent being domain generalization [56, 31, 53, 99, 6, 38], domain adaptation [98, 82, 51, 90, 91, 81], robust learning [42, 69, 94], federated learning [54, 36, 44, 97, 45], and OOD detection [41, 32, 92, 84, 49]. These paradigms are distinguished chiefly by their operating assumptions: domain adaptation presupposes access to unlabeled data drawn from the test set, whereas domain generalization leans upon environment labels furnished during training. Federated learning, for its part, embraces a collaborative learning framework and grapples with constraints such as communication efficiency and privacy that arise when data are dispersed across multiple environments, while OOD detection concerns itself with the identification of samples that depart markedly from the training distribution. A goal common to many of these methods is the recovery of invariant relationships. Recent work, however, intimates that inductive biases beyond invariance alone are requisite for improved generalization [46, 72, 93]. In particular, Ye et al. [93] contend that the invariance of features, while necessary,

does not suffice for generalization, and accordingly emphasize the *informativeness* of features as an additional desideratum. We likewise interrogate the joint role of invariance and informativeness in OOD generalization.

Proxy variable adjustment: For the regime in which confounding variables are observed during training yet become unobserved at test time, Landeiro and Culotta [40] advocate adjusting for confounding shifts as a means of improving classifier performance. Building upon proxy-based adjustment methods originally devised for causal effect estimation [55, 39], a line of recent domain adaptation methods recasts the problem of unknown distribution shifts as one of causal effect identification [4, 85]. More recently still, OOD generalization under hidden confounding shift has been examined under the assumption of overlapping confounder support [62]. In contradistinction to these approaches, our contribution is to *explain the mechanism by which* proxy variables enhance OOD generalization.

All variable models vs causal models. More recently, Nastl and Hardt [57] advanced a benchmark study in which covariates are partitioned into four categories: causal (conservatively delineated), arguably causal, anti-causal, and residual spurious covariates. Their central finding is that, across 16 benchmark datasets, models exploiting the full set of covariates Pareto-dominate those confined to the causal or arguably causal subsets, and do so on both ID and OOD data. Theoretical work accounting for this behavior remains scarce, and it is precisely such an account that we set out to provide, marshalling concrete scenarios and arguments to explain their experimental findings. In the linear regime, anchor regression [68] furnishes a framework that interpolates between two estimation paradigms: models that admit all observed covariates and models that restrict attention to causal covariates alone. Our aim is to elucidate the effect of admitting further, not-necessarily-causal covariates under a hidden confounding shift. Eastwood et al. [21] establish that unstable covariates can amplify performance when they carry information pertaining to the label, provided they are conditionally independent of the stable covariates given the label, and they propose to correct the distribution shift by inspecting the test domain in the absence of labels. Even so, when their method is brought to bear on a medical real-world dataset not purpose-built for this problem [8], ERM persists as a competitive alternative, in consonance with the findings of Nastl and Hardt [57]. This observation reflects the broader principle that, under well-specified covariate shifts, maximum likelihood estimation (MLE) attains minimax optimality for OOD generalization [26]. Real-world settings, however, are seldom well-specified, precisely on account of hidden confounding shift—the central preoccupation of the present work.

3 Manifestations of hidden confounding shift

In this section, we furnish the requisite background on hidden confounding shift and, by means of an example concerning the causal effect identification of observed covariates upon the outcome, motivate the imperative of addressing it.

Types of distribution shifts. For covariates $\mathbf{X}$ and target Y , several distinct distribution shifts may be observed between two environments e and e' , as enumerated in Table 1. Such shifts are ordinarily traceable to a shift in the distribution $\mathbb{P}(U)$ of an unobserved covariate U that causes $\mathbf{X}$, or Y , or indeed both (see Figure 1). Because the observed distributions involving only $\mathbf{X}$ and Y are functionally tied to $\mathbb{P}(U)$, any perturbation of the latter propagates to the former. To illustrate, one may write $\mathbb{P}(\mathbf{X}) = \sum_U \mathbb{P}(U)\mathbb{P}(\mathbf{X} | U)$; consequently, whenever $\mathbb{P}^e(U) \neq \mathbb{P}^{e'}(U)$ and $U \rightarrow \mathbf{X}$, it follows that $\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$. Strictly analogous reasoning governs the induced shifts of $\mathbb{P}(Y)$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y | \mathbf{X})$, $\mathbb{P}(\mathbf{X}, Y)$.

A number of existing methods for OOD generalization posit particular causal relationships among $U, \mathbf{X}, Y$ that in turn guarantee specific invariances. When $U \rightarrow Y \rightarrow \mathbf{X}$ (Figure 1 (a)), for instance, the factorization $\mathbb{P}(\mathbf{X}, Y) = \mathbb{P}(Y)\mathbb{P}(\mathbf{X} | Y)$ implies that any

Table 1: A synopsis of the several distribution shifts.

Type of Shift	Mathematical Expression
Label	$\mathbb{P}^e(Y) \neq \mathbb{P}^{e'}(Y)$
Covariate	$\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$
Conditional Covariate	$\mathbb{P}^e(\mathbf{X} Y) \neq \mathbb{P}^{e'}(\mathbf{X} Y)$
Concept	$\mathbb{P}^e(Y \mathbf{X}) \neq \mathbb{P}^{e'}(Y \mathbf{X})$
Dataset	$\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$

$U \rightarrow \mathbf{X}$; consequently, whenever $\mathbb{P}^e(U) \neq \mathbb{P}^{e'}(U)$

and $U \rightarrow \mathbf{X}$, it follows that $\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$. Strictly analogous reasoning governs the induced shifts of $\mathbb{P}(Y)$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y | \mathbf{X})$, $\mathbb{P}(\mathbf{X}, Y)$.

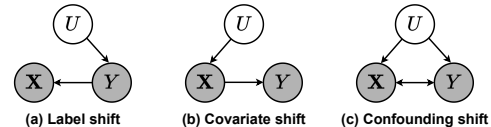


Figure 1: Causal graphs underlying distribution shifts.

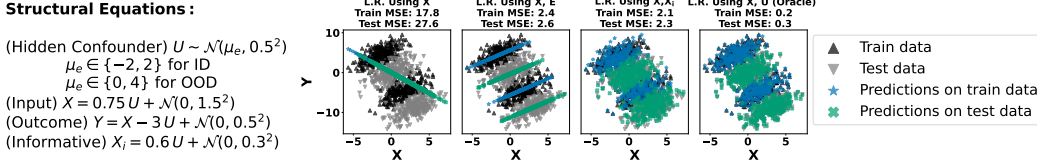


Figure 2: We evaluate four linear regression (L.R.) models within an OOD setting characterized by hidden confounding shifts and minimal environment overlap (i.e., distant μ_e). (i) A model trained on X alone learns an erroneous relationship with Y , thereby illustrating Simpson’s paradox. (ii) Recourse to environment-specific summary statistics of X , denoted E , recovers the correct relationship yet remains constrained in representational power. (iii) The use of an informative covariate X_i for U improves OOD generalization. (iv) The oracle model is trained on both X and U .

153 shift $\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$ is attributable to
154 label shift—that is, $\mathbb{P}^e(Y) \neq \mathbb{P}^{e'}(Y)$ —while the conditional covariate distribution nonetheless remains invariant, $\mathbb{P}^e(\mathbf{X} | Y) = \mathbb{P}^{e'}(\mathbf{X} | Y)$, by virtue of the identity $\mathbb{P}(\mathbf{X}|Y, U) = \mathbb{P}(\mathbf{X}|Y)$ [89, 83, 25, 5, 47, 96, 18, 79]. By parallel reasoning, when $U \rightarrow \mathbf{X} \rightarrow Y$ (Figure 1
157 (b)), a shift $\mathbb{P}^e(\mathbf{X}, Y) \neq \mathbb{P}^{e'}(\mathbf{X}, Y)$ arises from covariate shift, $\mathbb{P}^e(\mathbf{X}) \neq \mathbb{P}^{e'}(\mathbf{X})$, whereas the conditional distribution $\mathbb{P}(Y | \mathbf{X})$ holds invariant, $\mathbb{P}^e(Y | \mathbf{X}) = \mathbb{P}^{e'}(Y | \mathbf{X})$ [70, 16, 80, 11, 28]. Such invariances, however, are apt to fail in a great many scenarios, since U typically causes both $\mathbf{X}$ and Y (Figure 1 (c)) [50, 4, 40, 85, 62, 65, 64]. In this configuration, a shift in $\mathbb{P}(U)$
161 between environments may precipitate label shift, covariate shift, conditional covariate shift, and concept shift all at once. Confounding shift nonetheless induces an invariance of its own—namely
162 $\mathbb{P}^e(\mathbf{X}, Y | U) = \mathbb{P}^{e'}(\mathbf{X}, Y | U)$, designated *stable confounding shift* [85, 4]. Because U is unobserved, this invariance is, however, inaccessible from the observed data.

165 **Information-theoretic measures:** Following Federici et al. [22], we quantify distribution shifts
166 by recourse to mutual information. Concretely, we employ the quantities $I(\mathbf{X}; E)$, $I(Y; E)$,
167 $I(\mathbf{X}; E|Y)$, $I(Y; E|\mathbf{X})$, $I(\mathbf{X}, Y; E)$ —with E denoting the environment variable—to gauge, re-
168 spectively, the shifts in $\mathbb{P}(\mathbf{X})$, $\mathbb{P}(Y)$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y | \mathbf{X})$, $\mathbb{P}(\mathbf{X}, Y)$ across environments. As
169 noted above, existing methods for OOD generalization commonly presuppose either $I(\mathbf{X}; E | Y) = 0$,
170 via the label shift assumption, or $I(Y; E | \mathbf{X}) = 0$, via the covariate shift assumption. Under hidden
171 confounding shifts, however, both quantities may in general be nonzero.

172 **Challenges with hidden confounding shift:** To render concrete the challenges that hidden con-
173 founding shift poses for OOD generalization, we present the experiment depicted in Figure 2. Here
174 the outcome Y is governed by X together with a hidden confounder U , and environment-specific
175 perturbations in the mean of U , denoted μ_e , give rise to distinct environments. A linear regression
176 model trained exclusively on the observed covariate X infers an erroneous relationship between X
177 and Y , as the first subplot makes plain—an instance of Simpson’s paradox [73], in which Y increases
178 with X within each individual environment even as the model apprehends precisely the opposite trend.
179 Methods expressly engineered for OOD generalization, including IRM [6], VREX [38], and Group
180 DRO [69], exhibit an analogous pathology. For such scenarios, we establish theoretically in § 4 that
181 the optimal generalization strategy consists in learning environment-specific relationships, which
182 affords recovery of the correct relationship between X and Y . The second subplot bears this out:
183 environment-specific summary statistics of X —its mean, standard deviation, and quantiles—assist in
184 uncovering the true X – Y relationship. This maneuver mirrors the *backdoor adjustment* criterion of
185 causal effect estimation [58], wherein environment-specific information serves as a proxy through
186 which to adjust for confounders. As the same subplot intimates, however, the environment-specific
187 information encoded in observed covariates alone may prove insufficient, owing to its circumscribed
188 representational capacity. Yet, as we demonstrate in § 4.3, additional informative covariates acting as
189 proxies for the unobserved confounders can improve generalization performance still further, as the
190 third subplot attests. For the sake of comparison, we furnish an oracle model trained on both X and
191 U , which realizes the best attainable mean squared error.

192 4 Impact of hidden confounding shifts on OOD generalization

193 In this section, we undertake a theoretical study of the manifold ways in which hidden confounding
194 shifts bear upon OOD generalization. Let $\hat{Y} = (f \circ \phi)(\mathbf{X})$ denote the label predicted for an input
195 $\mathbf{X}$, where f is a classifier and ϕ a feature extractor or transformation function; further, let $H(X) =$

196 $-\mathbb{E}_X[\log(\mathbb{P}(X))]$ denote entropy and $I(X; Y) = \mathbb{E}_{X,Y}[\log \frac{\mathbb{P}(X,Y)}{\mathbb{P}(X)\mathbb{P}(Y)}]$ mutual information. The risk
 197 incurred by the predictor $(f \circ \phi)$ in an environment e is defined as $\mathcal{R}^e(f \circ \phi) = \mathbb{E}_{(\mathbf{X}, Y) \sim \mathbb{P}_{\mathbf{X}, Y}^e}[\ell((f \circ \phi)(\mathbf{X}), Y)]$, with $\ell(\cdot, \cdot)$ a loss function. The overarching objective of OOD generalization is the
 198 acquisition of a predictor $(f \circ \phi)$ that performs well on both ID and OOD data, and to this end a
 199 variety of objectives has been entertained in the literature. Robust optimization-based methods [7,
 200 9, 19, 75, 69] seek to minimize the worst-case risk across all training environments, $\mathcal{R}^{\text{Rob}}(f \circ \phi) =$
 201 $\max_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi)$, where $\mathcal{E}_{tr}$ denotes the set of training environments. Variance risk extrapolation
 202 (VREX) [38] aims to minimize $\mathcal{R}^{\text{VREX}}(f \circ \phi) = \sum_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi) + \beta \times \text{Variance}(\{\mathcal{R}^1(f \circ \phi)$
 203 $\phi), \dots, \mathcal{R}^{|\mathcal{E}_{tr}|}(f \circ \phi)\})$, in which β is a hyperparameter. Invariant risk minimization (IRM) [6]
 204 minimizes $\sum_{e \in \mathcal{E}_{tr}} \mathcal{R}^e(f \circ \phi)$ subject to the constraint that f be a simultaneously optimal classifier
 205 across all environments. Methods grounded in empirical risk minimization (ERM), by contrast,
 206 simply pool the data drawn from all training environments and minimize the empirical risk over the
 207 pooled data [6, 38]; that is, ERM minimizes $\mathcal{R}^{\text{ERM}}(f \circ \phi) = \sum_{e \in \mathcal{E}_{tr}} \sum_{(\mathbf{X}, Y) \in \mathcal{D}^e} \ell((f \circ \phi)(\mathbf{X}), Y)$,
 208 where $\mathcal{D}^e$ denotes the data observed in environment e .
 209

210 In order to apprehend how hidden confounding shift impinges upon traditional objective functions, we
 211 scrutinize several facets of the maximization of predictive information, $I(Y; \hat{Y})$ with $\hat{Y} = (f \circ \phi)(\mathbf{X})$.
 212 Since $I(Y; \hat{Y})$ quantifies the degree to which the prediction $\hat{Y}$ is informative about the true label Y ,
 213 it constitutes a natural and meaningful objective across a broad class of machine learning tasks. We
 214 commence by delineating two cardinal properties of $\phi(\mathbf{X})$ —*informativeness* and *invariance*—each
 215 of which is indispensable to OOD generalization [93].

216 **Definition 4.1** (Informativeness and Conditional Informativeness). *The informativeness of fea-*
 217 *tures $\phi(\mathbf{X})$ for predicting Y is defined as $\text{INF}(\phi(\mathbf{X}), Y) = I(\phi(\mathbf{X}); Y)$. The conditional informa-*
 218 *tiveness of features $\phi(\mathbf{X})$ for predicting Y conditioned on environment variable E is defined as*
 219 *$\text{CINF}(\phi(\mathbf{X}), Y, E) = I(\phi(\mathbf{X}); Y|E)$.*

220 Conditional informativeness measures the information that $\phi(\mathbf{X})$ conveys about Y within each
 221 environment. Minimizing CINF entails a loss of information, whereas maximizing it may prove
 222 undesirable in applications such as algorithmic fairness, since $\phi(\mathbf{X})$ might there exploit sensitive or
 223 biased information residing within environments and thereby yield unfair predictions. Although such
 224 biased representations can be forestalled by minimizing $I(\phi(\mathbf{X}); E)$, doing so may erode predictive
 225 performance, as we demonstrate in § 4.2. Confining attention to the setting in which no sensitive
 226 information attaches to E , we adopt—and subsequently substantiate in § 4.1—the position that
 227 maximizing CINF can improve generalization performance.

228 **Definition 4.2** (Variation and Invariance). *For a given label Y , the variation in the features $\phi(\mathbf{X})$*
 229 *across environments E is defined as $\text{VAR}(\phi(\mathbf{X}), Y, E) = I(\phi(\mathbf{X}); E|Y)$. For a given label Y , the*
 230 *invariance of the features $\phi(\mathbf{X})$ across environments E is defined as the negative of the variation i.e.,*
 231 *$\text{INV}(\phi(\mathbf{X}), Y, E) = -\text{VAR}(\phi(\mathbf{X}), Y, E) = -I(\phi(\mathbf{X}); E|Y)$*

232 Invariance quantifies the extent to which the representation $\phi(\mathbf{X})$ remains consistent across differing
 233 environments for a given label Y . To minimize invariance is to invite overfitting, inasmuch as
 234 environment-specific information is thereby preserved within $\phi(\mathbf{X})$; to maximize it, conversely, is
 235 to expunge environment-specific dependencies and so to promote invariant learning. Extending the
 236 information-theoretic measures introduced in § 3, we deploy $I(\phi(\mathbf{X}); E)$ and $I(Y; E | \phi(\mathbf{X}))$ to
 237 quantify *feature shift* and *concept shift*, respectively. As context dictates, the term *concept shift* shall
 238 denote either $I(Y; E | \mathbf{X})$ or $I(Y; E | \phi(\mathbf{X}))$.

239 4.1 A general decomposition of predictive information

240 We preface the analysis with a handful of causal-graph preliminaries. A causal graph $\mathcal{G}$ comprises
 241 nodes, which represent random variables, together with directed edges, which encode direct causal
 242 influences between nodes. A *path* between two nodes X_i and X_j is a sequence of distinct nodes joined
 243 by edges; a *directed path* from X_i to X_j with $i < j$ is one along which every arrow points toward
 244 X_j , namely $X_i \rightarrow X_{i+1} \rightarrow \dots \rightarrow X_{j-1} \rightarrow X_j$. Within such a directed path, X_i is termed the
 245 *parent* of X_{i+1} , X_{i+1} the *child* of X_i , X_i an *ancestor* of X_j , and X_j a *descendant* of X_i . Every path
 246 decomposes into three fundamental structures: a *chain* $X_i \rightarrow X_j \rightarrow X_k$, a *fork* $X_i \leftarrow X_j \rightarrow X_k$,
 247 and a *collider* $X_i \rightarrow X_j \leftarrow X_k$. In chains and forks alike, X_i and X_k are marginally dependent
 248 yet are rendered conditionally independent once one conditions upon the intermediate node X_j , i.e.
 249 $X_i \not\perp\!\!\!\perp X_k$ and $X_i \perp\!\!\!\perp X_k | X_j$. In a collider, by contrast, X_i and X_k are marginally independent but

become conditionally dependent upon conditioning on X_j or any of its descendants, i.e. $X_i \perp\!\!\!\perp X_k$ and $X_i \not\perp\!\!\!\perp X_k \mid X_j$. A path between X_i and X_k is said to be *blocked* by a conditioning set $\mathcal{S}$ if and only if either (i) the path contains a chain or fork whose middle node lies in $\mathcal{S}$, or (ii) it contains a collider such that neither the collider nor any of its descendants belongs to $\mathcal{S}$. Should every path from X_i to X_k be blocked by $\mathcal{S}$, then $X_i \perp\!\!\!\perp X_k \mid \mathcal{S}$; a path that is not blocked is said to be open.

Consider now the predictive information $I(Y; \hat{Y})$, wherein the predictions $\hat{Y} = (f \circ \phi)(\mathbf{X})$ rest upon a learned representation $\phi(\mathbf{X})$. We model the causal relationships obtaining among $\mathbf{X}, Y, U, \phi(\mathbf{X}), \hat{Y}$ and E as portrayed in Figure 3, where U figures as a hidden confounding variable. Conditioned upon either $\mathbf{X}$ or $\phi(\mathbf{X})$, the prediction $\hat{Y}$ is redundant for the purpose of reasoning about Y ; that is, $\hat{Y} \perp\!\!\!\perp Y \mid \mathbf{X}$ and $\hat{Y} \perp\!\!\!\perp Y \mid \phi(\mathbf{X})$. The variable E denotes an environment variable that registers shifts in $\mathbb{P}(U)$ —which is to say that between any two environments there subsists a shift in $\mathbb{P}(U)$. We first set forth a more general decomposition of predictive information that abstracts away from the specific manner in which U influences $\mathbf{X}, Y$; a formal treatment of U 's influence upon $\mathbf{X}$ and Y is deferred to § 4.2.

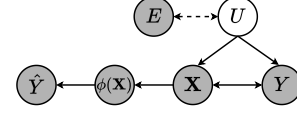


Figure 3: Bi-directed arrow between $\mathbf{X}$ and Y indicate that some covariates of $\mathbf{X}$ can cause Y , and some may be caused by Y .

Proposition 4.1. For a covariate vector $\mathbf{X}$, label Y , with causal structure $\mathbf{X} \leftrightarrow Y$, i.e., some covariates cause Y and some covariates are caused by Y , environment variable E , a feature extractor ϕ , and prediction $\hat{Y}$, the predictive information $I(Y; \hat{Y})$ is decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y | E)}^{\text{Cond. informativ.}} - \overbrace{\frac{I(\phi(\mathbf{X}); E | Y)}{2}}^{\text{Variation}} + \overbrace{\frac{I(Y; E)}{2}}^{\text{Label shift}} + \overbrace{\frac{I(\phi(\mathbf{X}); E)}{2}}^{\text{Feature shift}} - \overbrace{\frac{I(Y; E | \phi(\mathbf{X}))}{2}}^{\text{Concept shift}} - \overbrace{I(\phi(\mathbf{X}); Y | \hat{Y})}^{\text{Residual}} \quad (1)$$

where $I(\phi(\mathbf{X}); Y | \hat{Y})$ is the residual information in $\phi(\mathbf{X})$ for inferring Y that is not captured by the prediction $\hat{Y}$. The decomposition above also holds when $\phi(\mathbf{X})$ is replaced with $\mathbf{X}$.

The proofs are relegated to Appendix § A. Existing methods may be construed as procedures that explicitly minimize or maximize particular terms of Equation (1). IRM [6], for instance, enforces that a fixed classifier f remain invariant across environments—a stipulation directly tantamount to maximizing $-I(\phi(\mathbf{X}); E | Y)$, since, were $\phi(\mathbf{X})$ to carry surplus environment-specific information conditional on Y , the fixed classifier could no longer be optimal in every environment. DANN [23] and the independence criterion for fair classification [22], for their part, endeavor to purge environment-specific information from $\phi(\mathbf{X})$ by minimizing $I(\phi(\mathbf{X}); E)$, and, in the setting of domain adaptation, certain properties of features are learned so as to be invariant to domains [82, 90]. Salutary though the minimization of $I(\phi(\mathbf{X}); E)$ may be for fair classification, it stands to degrade predictive performance [98, 33, 22]—a liability legible in Equation (1), where a reduction of $I(\phi(\mathbf{X}); E)$ can in turn reduce $I(Y; \hat{Y})$. In CDAN [51], the objective is to secure $\mathbb{P}^e(\phi(\mathbf{X}), Y) = \mathbb{P}^{e'}(\phi(\mathbf{X}), Y)$; enforcing this equality renders E independent of the pair $(\phi(\mathbf{X}), Y)$, whence $\mathbb{P}(Y | \phi(\mathbf{X}), E) = \mathbb{P}(Y | \phi(\mathbf{X})) \implies I(Y; E | \phi(\mathbf{X})) = 0$. Thus, although CDAN does not explicitly incorporate $I(Y; E | \phi(\mathbf{X}))$ within its loss, its joint-distribution alignment objective effectively drives this conditional mutual information to zero. Finally, while $I(Y; E)$ is a constant, there exist methods engineered to be robust to shifts in the distribution of labels [69, 95].

We recall that Equation (1) is derived without explicit regard to the manner in which U influences $\mathbf{X}, Y$. Knowledge of that influence can, however, further guide the effective treatment of the terms in (1). Specifically, if $U \rightarrow Y, U \not\rightarrow \mathbf{X}, Y \rightarrow \mathbf{X}$, then $I(\mathbf{X}; E | Y) = 0$, which intimates that the minimization of $I(\phi(\mathbf{X}); E | Y)$ is a principled objective; symmetrically, if $U \not\rightarrow Y, U \rightarrow \mathbf{X}, \mathbf{X} \rightarrow Y$, then $I(Y; E | \mathbf{X}) = 0$, thereby motivating the minimization of $I(Y; E | \phi(\mathbf{X}))$. When, however, $U \rightarrow Y$ and $U \rightarrow \mathbf{X}$, the interplay among the terms of Equation (1) grows non-trivial, and it ceases to be evident what an ideal strategy for addressing them might be. It is precisely this question that we resolve in the section that follows.

4.2 Predictive information under hidden confounding shift

To apprehend how a hidden confounding variable U that causes both $\mathbf{X}$ and Y bears upon the predictive information, we examine *two special cases* of the relationship between $\mathbf{X}$ and Y : (i) $\mathbf{X} \rightarrow Y$ and (ii) $Y \rightarrow \mathbf{X}$. In the case $\mathbf{X} \rightarrow Y$, the inequalities in (2) obtain, for conditioning on

300 Y opens the path $\phi(\mathbf{X}) \leftarrow \mathbf{X} \rightarrow Y \leftarrow U \leftarrow E$ from $\phi(\mathbf{X})$ to E at the collider node Y , thereby
 301 admitting additional information flow from $\phi(\mathbf{X})$ to E by way of Y . Conditioning on $\phi(\mathbf{X})$, by
 302 contrast, partially blocks the flow of information from Y to E at the node $\mathbf{X}$, provided that $\phi(\mathbf{X})$
 303 encodes at least some information about $\mathbf{X}$.

$$(i) \overbrace{I(\phi(\mathbf{X}); E|Y)}^{\text{Variation}} \geq \overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}} \quad (ii) \overbrace{I(Y; E)}^{\text{Label shift}} \geq \overbrace{I(Y; E|\phi(\mathbf{X}))}^{\text{Concept shift}} \quad (2)$$

304 By parallel reasoning, in the case $Y \rightarrow \mathbf{X}$ the inequalities in (3) follow, for conditioning on Y blocks
 305 the flow of information from $\phi(\mathbf{X})$ to E at the node Y , while conditioning on $\phi(\mathbf{X})$ opens the path
 306 $Y \rightarrow \mathbf{X} \leftarrow U \leftarrow E$ from Y to E at the node $\mathbf{X}$ — $\phi(\mathbf{X})$ being a child of $\mathbf{X}$, and conditioning on the
 307 child of a collider serving to open a path through that collider [58] (recall the preliminaries of § 4.1).

$$(i) \overbrace{I(\phi(\mathbf{X}); E)}^{\text{Feature shift}} \geq \overbrace{I(\phi(\mathbf{X}); E|Y)}^{\text{Variation}} \quad (ii) \overbrace{I(Y; E|\phi(\mathbf{X}))}^{\text{Concept shift}} \geq \overbrace{I(Y; E)}^{\text{Label shift}} \quad (3)$$

308 Invoking the inequalities in (2) and (3), we refine the predictive information decomposition as follows.
 309

310 **Proposition 4.2.** *For a covariate vector $\mathbf{X}$, label Y , an environment variable E , a feature extractor*
 311 *ϕ , the prediction $\hat{Y}$, and an unobserved confounding variable U that cause both $\mathbf{X}$ and Y , and if*
 312 *either (i) $\mathbf{X} \rightarrow Y$ or (ii) $Y \rightarrow \mathbf{X}$, the predictive information $I(Y; \hat{Y})$ can be decomposed as follows:*

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativeness}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (4)$$

313 In Proposition 4.2, we treat the two cases— $\mathbf{X} \rightarrow Y$ and $Y \rightarrow \mathbf{X}$ —separately. The causal relationship
 314 $\mathbf{X} \rightarrow Y$ is nonetheless the more prevalent of the two in real-world tabular prediction tasks: 11 of
 315 the 16 datasets considered in recent benchmark studies [57, 50, 24] conform to the causal structure
 316 $\mathbf{X} \rightarrow Y$, and even within the remaining 5 the covariates caused by Y are far outnumbered by those
 317 that cause Y . We set forth the principal takeaway from Proposition 4.2 below.

In contradistinction to the decomposition in (1), the decomposition in (4)—obtained for the special cases (i) $U \rightarrow \mathbf{X}, U \rightarrow Y, \mathbf{X} \rightarrow Y$ and (ii) $U \rightarrow \mathbf{X}, U \rightarrow Y, Y \rightarrow \mathbf{X}$ —is purged of the terms *variation, label shift, feature shift, and concept shift*. In other words, under hidden confounding shift, whenever either $\mathbf{X} \rightarrow Y$ or $Y \rightarrow \mathbf{X}$, the predictive information stands to be maximized by maximizing the difference *conditional informativeness* – *residual*.

318
 319 **Maximizing conditional informativeness:** In very recent work, Prashant et al. [62] put forward a
 320 mixture-of-experts (MoE) model for OOD generalization under hidden confounding shift, in which
 321 each expert is associated with a particular hidden confounder assignment—that is, each expert
 322 concentrates on maximizing performance within the environment corresponding to a specific value of
 323 the hidden confounder. The learning of MoE models thus coincides with the objective of maximizing
 324 the conditional informativeness $I(\phi(\mathbf{X}); Y|E)$, and Equation (4) accordingly furnishes theoretical
 325 justification for the deployment of MoE-type models. Beyond lending support to such models,
 326 Equation (4) further throws into relief the need for methods capable of operating under more general
 327 confounding shift regimes: the method advanced by Prashant et al. [62], for example, presupposes
 328 overlapping confounding support, $\text{supp}(\mathbb{P}(U|\mathcal{E}_{te})) \subseteq \text{supp}(\mathbb{P}(U|\mathcal{E}_{tr}))$, together with proxy variables
 329 that are discrete-valued.

The results in Figure 2 demonstrate that, in a simple setting devoid of confounding overlap and endowed with a continuous-valued proxy, a linear regression model, once supplied with environment-specific information, succeeds in recovering the correct causal relationship between observed covariates and the target across environments. This constitutes a motivating example that underscores the need for OOD generalization methods which retain their effectiveness even where confounding support is limited and the confounder is continuous-valued.

330
 331 **Minimizing residual:** The residual term $I(\phi(\mathbf{X}); Y | \hat{Y})$ quantifies the surplus information that
 332 $\phi(\mathbf{X})$ conveys about the true label Y beyond that already embodied in the prediction $\hat{Y}$, and may

be written as $I(\phi(\mathbf{X}); Y | \hat{Y}) = H(Y | \hat{Y}) - H(Y | \phi(\mathbf{X}), \hat{Y})$. The conditional entropy $H(Y | \hat{Y})$ stands in relation to the expected cross-entropy loss through the identity $\mathbb{E}_{Y, \hat{Y}}[\ell_{\text{CE}}(Y, \hat{Y})] = H(Y | \hat{Y}) + \text{calibration error}$ [12, 10, 29], wherein $H(Y | \hat{Y})$ is known as the *refinement error* and measures the model’s capacity to discriminate among classes. The minimization of the residual term $I(\phi(\mathbf{X}); Y | \hat{Y})$ thus admits two complementary strategies. The first is to minimize $I(\phi(\mathbf{X}); Y | \hat{Y})$ directly, by rendering $\phi(\mathbf{X})$ conditionally independent of the target Y given $\hat{Y}$, i.e., $Y \perp\!\!\!\perp \phi(\mathbf{X}) | \hat{Y}$. The second is to minimize the refinement error $H(Y | \hat{Y})$ —for instance, through training procedures that first minimize the refinement error, so as to secure accurate predictions, before turning to the calibration error [10].

4.3 Impact of additional informative variables

In § 4.2, we set forth the optimal strategies for generalization in the regime where the confounder goes unobserved. In practice, one is on occasion afforded access to proxies for the hidden confounding variable [4, 85, 62], which may then be enlisted as surrogates for it. We now investigate the effect that the addition of such non-causal yet informative covariates exerts upon predictive information.

Definition 4.3 (Informative Covariates). *A set of covariates $\mathbf{X}_I$ that are not causally related to Y i.e., neither ancestors nor descendants of Y are in $\mathbf{X}_I$, are said to be informative to Y if $\mathbf{X}_I$ and Y are not independent of each other given other causally related covariates $\mathbf{X}$ and E i.e., $Y \not\perp\!\!\!\perp \mathbf{X}_I | \mathbf{X}, E$.*

Since $U \rightarrow Y$, any covariate informative of U is likewise informative of Y , and conversely. It follows from the predictive information decomposition in (1) that the maximization of $I(Y; \hat{Y})$ demands the minimization of the concept shift $I(Y; E | \phi(\mathbf{X}))$. Liu et al. [50], on the strength of their empirical analysis, suggest that gathering additional covariates $\mathbf{X}_I$ such that $\mathbb{P}(Y | \mathbf{X}, \mathbf{X}_I)$ is rendered more stable across environments—that is, such that concept shift is diminished—improves OOD test accuracy. We establish theoretically that the exploitation of further informative variables assists in maximizing predictive information, by way of maximizing or minimizing particular terms in (1).

Proposition 4.3. *If $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and if $\mathbf{X}_I$ are informative variables such that $Y \not\perp\!\!\!\perp \mathbf{X}_I | \mathbf{X}, E$, then we have the following inequalities:*

$$\begin{aligned} (i) \ I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y | E) &> I(\phi_1(\mathbf{X}); Y | E) \quad (ii) \ I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E) \\ (iii) \ I(Y; E | \phi_2(\{\mathbf{X} \cup \mathbf{X}_I\})) &< I(Y; E | \phi_1(\mathbf{X})) \quad (iv) \ I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E | Y) > I(\phi_1(\mathbf{X}); E | Y) \end{aligned} \quad (5)$$

Proposition 4.3 establishes that the addition of informative covariates augments conditional informativeness (5.i) and feature shift (5.ii) while attenuating concept shift (5.iii). The same addition, however, simultaneously amplifies variation (5.iv), a consequence that may in turn call for dedicated strategies by which to hold the variation in check.

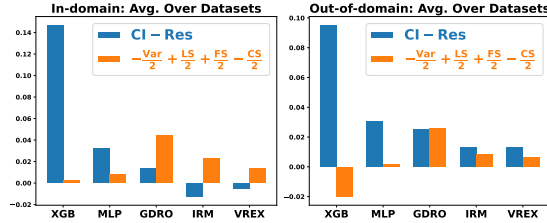
5 Experimental results

We conduct experiments upon both real-world and synthetic datasets in order to analyze: (i) the presence of hidden confounding shift within real-world data, (ii) the manner in which the constituent components of the decomposition in (1) bear upon OOD generalization under hidden confounding shifts, and (iii) the role played by informative covariates in improving generalization. We consider eight real-world tabular benchmark datasets—*Food stamps*, *Readmission*, *Income*, *Public coverage*, *Unemployment*, *Diabetes*, *Hypertension*, and *ASSISTments*—whose datasets and attendant domain splits are drawn from the *TableShift* benchmark [24]. Model performance is assessed by means of ID test and OOD test accuracies. Our experiments encompass two ERM-based methods, XGBoost (XGB) [15] and a multi-layer perceptron (MLP); two domain generalization methods, IRM [6] and VREX [38]; and one robust-learning method, Group DRO (GDRO) [69]. We report results averaged over 5 distinct random hyperparameter choices, as detailed in Appendix B. The influence of random seeds is statistically insignificant for these datasets [24]. Supplementary details of the experimental setup are furnished in Appendix § B, and results obtained under hyperparameter tuning are reported in Appendix C.

Hidden confounding shift: We recall that hidden confounding shifts induce observable shifts in the distributions $\mathbb{P}(\mathbf{X})$, $\mathbb{P}(\mathbf{X} | Y)$, $\mathbb{P}(Y)$, and $\mathbb{P}(Y | \mathbf{X})$. The results in Table 2 reveal that label shift, covariate shift, conditional covariate shift, and concept shift are all in evidence across the real-world datasets, which is itself indicative of the presence of hidden confounding shifts. For the qualitative

Table 2: Quantification of distribution shifts. The mean $\pm$ standard deviation is computed over 10 random subsets of 40,000 samples apiece. Statistical significance against a null mean of zero is adjudged via one-sample t-tests ($\alpha = 0.05$), which confirm that every measure differs significantly from zero (p-value ≈ 0).

Dataset	Conditional Covariate Shift $I(\mathbf{X}; E Y)$	Label shift $I(Y; E)$	Covariate Shift $I(\mathbf{X}; E)$	Concept shift $I(Y; E \mathbf{X})$
Readmission	0.107 ± 0.002	0.068 ± 0.002	0.097 ± 0.002	2.032 ± 0.000
Food stamps	0.126 ± 0.004	0.030 ± 0.003	0.108 ± 0.001	2.118 ± 0.001
Income	0.168 ± 0.002	0.075 ± 0.003	0.147 ± 0.001	2.059 ± 0.002
Public coverage	0.231 ± 0.002	0.412 ± 0.006	0.222 ± 0.002	1.945 ± 0.001
Unemployment	0.117 ± 0.001	0.019 ± 0.002	0.114 ± 0.002	2.010 ± 0.003
Diabetes	0.032 ± 0.002	0.048 ± 0.001	0.022 ± 0.002	2.132 ± 0.001
Hypertension	0.090 ± 0.002	0.183 ± 0.003	0.037 ± 0.001	1.883 ± 0.004
ASSISTments	0.293 ± 0.002	0.260 ± 0.001	0.306 ± 0.002	0.367 ± 0.004



Method	Avg. ID Test Acc.	Avg. OOD Test Acc.
XGB	85.22	77.00
MLP	83.64	73.29
GDRO	78.40	72.28
IRM	71.17	64.89
VREX	59.74	63.06

Figure 4: The difference *conditional informativeness* – *residual* depicted in the plots is positively correlated with the average ID and OOD test accuracy computed over the eight datasets tabulated to the right.

analysis, we query GPT-4o [1] to enumerate the potential hidden confounders attending several benchmark datasets [24, 57], with the ensuing results reported in Appendix § E. In the *food stamps* dataset, for example, unmeasured factors such as the *economic policies* peculiar to each state may bear upon both *household* income and *food stamp* reciprocity.

Conditional informativeness vs. accuracy: By Proposition 4.2, under hidden confounding shift the maximization of the difference *conditional informativeness* – *residual* is essential to the maximization of predictive information. Results on the first six datasets—those possessing more than a single training environment [57, 24]—enumerated in Table 2 are displayed in Figure 4, with dataset-wise results relegated to Appendix C. These results attest that the difference *conditional informativeness* – *residual* is positively correlated with both ID and OOD accuracy, XGB emerging as the best-performing model and, correspondingly, the one attaining the highest value of *conditional informativeness* – *residual*. As the dataset-wise results in Appendix C make plain, we observe that in a great many settings, on account of hidden confounding shifts, the bars rendered in orange hover near zero, precisely because the sum $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$ vanishes under hidden confounding shift (Proposition 4.2). The relative ID / OOD performance of the methods depicted in Figure 1 accords with results previously reported in the literature [57, 24].

Informative covariates vs. accuracy: We turn now to the question of how the inclusion of informative covariates conduces to superior OOD generalization. To this end, we adopt the covariate partitioning of [57], whereby all available covariates are marshalled into three nested subsets, *causal covariates* $\subseteq$ *arguably causal covariates* $\subseteq$ *all covariates* (see Appendix § B), and we evaluate how the terms in the decomposition in (1) affect ID and OOD performance as one passes from one covariate subset to a larger one. For this purpose we introduce what we designate the *sign consistency metric*, which operates as follows. Upon moving from one covariate set to a larger one, for any term in (1) bearing a positive coefficient—conditional informativeness, say—we tally how frequently its value rises in concert with an improvement in accuracy; for negatively weighted terms, such as concept shift, we tally how frequently its value falls as accuracy improves. The metric is then the fraction of observations in which a term’s change is aligned with its *beneficial* direction, and thereby captures how reliably that term conduces to improved generalization. The formal definition is provided in Appendix § B. Table 3 demonstrates that both conditional informativeness and concept shift exhibit high sign consistency with the observed accuracy gains. While the reduction of concept shift is of the first importance, XGB additionally elevates conditional informativeness beyond the levels attained by

Table 3: Across all methods, the sign consistency value is high for concept shift (CS). It is, however, the sign consistency of conditional informativeness (CI) that is decisive for generalization, per Proposition 4.2, and it is at this that XGB excels. Res: residual, Var: variation, FS: feature shift, C: causal, AC: arguably causal, A: all.

Method	Sign Consistency Metric ($\uparrow$)					ID-Test Accuracy ($\uparrow$)			OOD-Test Accuracy ($\uparrow$)		
	CI	Var	FS	CS	Res	C	AC	A	C	AC	A
XGB	0.89	0.61	0.36	0.89	0.17	81.24	84.84	85.22	69.92	76.72	77.00
MLP	0.75	0.64	0.28	0.89	0.33	80.51	81.93	83.64	69.70	73.84	73.29
GDRO	0.69	0.67	0.19	0.94	0.36	80.92	83.76	78.40	69.61	73.49	72.28
IRM	0.78	0.61	0.25	0.92	0.25	67.91	68.22	71.17	61.21	62.96	64.89
VREX	0.64	0.61	0.25	0.89	0.44	54.39	62.32	59.74	56.62	63.53	63.06

Mean Squared Error

Cond. Informativeness & Variation

Feature Shift & Concept Shift

● Linear Regression ▲ XGB ● GDRO ■ IRM ◆ VREX Cond. Informativeness — Variation - - - Feature Shift - · - Concept Shift

Figure 5: The introduction of additional proxy variables $\mathbf{X}_I$ of U that are informative of Y conduces to a reduction in MSE and to increases in conditional informativeness and feature shift, all the while diminishing concept shift.

competing methods, and so realizes further performance improvements. Dataset-specific results are presented in Appendix § C.

It bears noting that, in real-world datasets, it is frequently arduous to ascertain whether *all available covariates* in fact encompass *every relevant or sufficiently informative* variable. To subject our hypothesis—that informative covariates enhance generalization—to further scrutiny, we conduct experiments on synthetic data endowed with a known causal structure: $U \rightarrow X$, $U \rightarrow Y$, $X \rightarrow Y$, and $U \rightarrow \mathbf{X}_I$, where $\mathbf{X}_I$ denotes the informative covariates caused by the hidden confounder U . We observe that the inclusion of $\mathbf{X}_I$ occasions improvements in conditional informativeness and feature shift while diminishing concept shift, as displayed in Figure 5. Tellingly, the variation component remains uniformly zero, a consequence of the simplicity of the problem. The synthetic data generation process, together with additional results, is provided in Appendix § D.

6 Discussion, limitations, and future work

In sum, we decompose the predictive information between model outputs and true outcomes so as to lay bare the factors that circumscribe OOD generalization (Equation (1)). We contend that hidden confounding shifts—manifested in the alterations of covariate and outcome distributions across environments—are ineluctable, and that no finite set of observed covariates or proxies can wholly dissolve the ambiguity they induce. Because confounding entangles covariate, label, concept, and feature shifts (the inequalities in (2), (3)), any attempt to treat these in isolation is bound to prove ineffectual. We demonstrate that the learning of environment-specific input–output mappings is more efficacious than the imposition of a single invariant predictor (Equation (4)). By foregrounding the inevitability of hidden confounding and the necessity of confronting it head-on, our work furnishes a foundation for more robust OOD methods. The principal challenges that lie ahead include: (i) the development of algorithms that remain effective under limited confounder overlap; (ii) the quantification of the cost–accuracy trade-off entailed in acquiring informative covariates; (iii) the treatment of entangled shifts without recourse to untestable proxy assumptions; and (iv) the modeling of the ambiguity arising from unobserved confounders, so as to inspire novel OOD-robust paradigms.

References

- [1] Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altschmidt, Sam Altman, Shyamal Anadkat, et al. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.
- [2] Alekh Agarwal, Alina Beygelzimer, Miroslav Dudík, John Langford, and Hanna Wallach. A reductions approach to fair classification. In *International conference on machine learning*, pages 60–69. PMLR, 2018.

- 446 [3] Hana Ajakan, Pascal Germain, Hugo Larochelle, François Laviolette, and Mario Marchand.
447 Domain-adversarial neural networks. *arXiv preprint arXiv:1412.4446*, 2014.
- 448 [4] Ibrahim Alabdulmohsin, Nicole Chiou, Alexander D’Amour, Arthur Gretton, Sanmi Koyejo,
449 Matt J. Kusner, Stephen R. Pfohl, Olawale Salaudeen, Jessica Schrouff, and Katherine Tsai.
450 Adapting to latent subgroup shifts via concepts and proxies. In *Proceedings of The 26th*
451 *International Conference on Artificial Intelligence and Statistics*, volume 206 of *Proceedings*
452 *of Machine Learning Research*, pages 9637–9661. PMLR, 25–27 Apr 2023.
- 453 [5] Amr Alexandari, Anshul Kundaje, and Avanti Shrikumar. Maximum likelihood with bias-
454 corrected calibration is hard-to-beat at label shift adaptation. In *International Conference on*
455 *Machine Learning*, pages 222–232. PMLR, 2020.
- 456 [6] Martin Arjovsky, Léon Bottou, Ishaan Gulrajani, and David Lopez-Paz. Invariant risk mini-
457 mization. *arXiv preprint arXiv:1907.02893*, 2019.
- 458 [7] J Andrew Bagnell. Robust supervised learning. In *AAAI*, pages 714–719, 2005.
- 459 [8] Peter Bandi, Oscar Geessink, Quirine Manson, Marcory Van Dijk, Maschenka Balkenhol,
460 Meyke Hermesen, Babak Ehteshami Bejnordi, Byungjae Lee, Kyunghyun Paeng, Aoxiao
461 Zhong, Quanzheng Li, Farhad Ghazvinian Zanjani, Svitlana Zinger, Keisuke Fukuta, Daisuke
462 Komura, Vlado Ovtcharov, Shenghua Cheng, Shaoqun Zeng, Jeppe Thagaard, Anders B
463 Dahl, Huangjing Lin, Hao Chen, Ludwig Jacobsson, Martin Hedlund, Melih Cetin, Eren
464 Halici, Hunter Jackson, Richard Chen, Fabian Both, Jorg Franke, Heidi Kusters-Vandeveld,
465 Willem Vreuls, Peter Bult, Bram van Ginneken, Jeroen van der Laak, and Geert Litjens. From
466 detection of individual metastases to classification of lymph node status at the patient level:
467 The CAMELYON17 challenge. *IEEE Trans Med Imaging*, 38, 2019.
- 468 [9] Aharon Ben-Tal and Arkadi Nemirovski. Robust optimization—methodology and applications.
469 *Mathematical programming*, 92:453–480, 2002.
- 470 [10] Eugene Berta, David Holzmüller, Michael I Jordan, and Francis Bach. Rethinking early
471 stopping: Refine, then calibrate. *arXiv preprint arXiv:2501.19195*, 2025.
- 472 [11] Steffen Bickel, Michael Brückner, and Tobias Scheffer. Discriminative learning under covariate
473 shift. *Journal of Machine Learning Research*, 10(9), 2009.
- 474 [12] Jochen Bröcker. Reliability, sufficiency, and the decomposition of proper scores. *Quarterly*
475 *Journal of the Royal Meteorological Society: A journal of the atmospheric sciences, applied*
476 *meteorology and physical oceanography*, 135:1512–1519, 2009.
- 477 [13] Donald T Campbell and Julian C Stanley. *Experimental and quasi-experimental designs for*
478 *research*. Ravenio books, 2015.
- 479 [14] Raymond J Carroll, David Ruppert, Leonard A Stefanski, and Ciprian M Crainiceanu. *Mea-*
480 *surement error in nonlinear models: a modern perspective*. Chapman and Hall/CRC, 2006.
- 481 [15] Tianqi Chen and Carlos Guestrin. Xgboost: A scalable tree boosting system. In *Proceedings*
482 *of the 22nd acm sigkdd international conference on knowledge discovery and data mining*,
483 pages 785–794, 2016.
- 484 [16] Xiangli Chen, Mathew Monfort, Anqi Liu, and Brian D Ziebart. Robust covariate shift
485 regression. In *Artificial Intelligence and Statistics*, pages 1270–1279. PMLR, 2016.
- 486 [17] Mathieu Chevalley, Charlotte Bunne, Andreas Krause, and Stefan Bauer. Invariant causal
487 mechanisms through distribution matching. *arXiv preprint arXiv:2206.11646*, 2022.
- 488 [18] Marthinus Christoffel Du Plessis and Masashi Sugiyama. Semi-supervised learning of class
489 balance under class-prior change by distribution matching. *Neural Networks*, 50:110–119,
490 2014.
- 491 [19] John C Duchi, Peter W Glynn, and Hongseok Namkoong. Statistics of robust optimization:
492 A generalized empirical likelihood approach. *Mathematics of Operations Research*, 46(3):
493 946–969, 2021.

- 494 [20] Cynthia Dwork, Aaron Roth, et al. The algorithmic foundations of differential privacy.
495 *Foundations and Trends® in Theoretical Computer Science*, 9:211–407, 2014.
- 496 [21] Cian Eastwood, Shashank Singh, Andrei L Nicolicioiu, Marin Vlastelica Pogančić, Julius
497 von Kügelgen, and Bernhard Schölkopf. Spuriousity didn’t kill the classifier: Using invariant
498 predictions to harness spurious features. *Advances in Neural Information Processing Systems*,
499 2023.
- 500 [22] Marco Federici, Ryota Tomioka, and Patrick Forré. An information-theoretic approach to
501 distribution shifts. In A. Beygelzimer, Y. Dauphin, P. Liang, and J. Wortman Vaughan, editors,
502 *Advances in Neural Information Processing Systems*, 2021.
- 503 [23] Yaroslav Ganin, Evgeniya Ustinova, Hana Ajakan, Pascal Germain, Hugo Larochelle, François
504 Laviolette, Mario March, and Victor Lempitsky. Domain-adversarial training of neural
505 networks. *Journal of machine learning research*, 17:1–35, 2016.
- 506 [24] Joshua P Gardner, Zoran Popovi, and Ludwig Schmidt. Benchmarking distribution shift in
507 tabular data with tableshift. In *Thirty-seventh Conference on Neural Information Processing*
508 *Systems Datasets and Benchmarks Track*, 2023.
- 509 [25] Saurabh Garg, Yifan Wu, Sivaraman Balakrishnan, and Zachary Lipton. A unified view of
510 label shift estimation. *Advances in Neural Information Processing Systems*, 33:3290–3300,
511 2020.
- 512 [26] Jiawei Ge, Shange Tang, Jianqing Fan, Cong Ma, and Chi Jin. Maximum likelihood estimation
513 is all you need for well-specified covariate shift. In *The Twelfth International Conference on*
514 *Learning Representations*, 2024.
- 515 [27] Yury Gorishniy, Ivan Rubachev, Valentin Khrulkov, and Artem Babenko. Revisiting deep
516 learning models for tabular data. *Advances in neural information processing systems*, 34:
517 18932–18943, 2021.
- 518 [28] Arthur Gretton, Alex Smola, Jiayuan Huang, Marcel Schmittfull, Karsten Borgwardt, Bernhard
519 Schölkopf, et al. Covariate shift by kernel mean matching. *Dataset shift in machine learning*,
520 3(4):5, 2009.
- 521 [29] Sebastian Gregor Gruber and Francis R. Bach. Optimizing estimators of squared calibration
522 errors in classification. *Transactions on Machine Learning Research*, 2025.
- 523 [30] Ishaan Gulrajani and David Lopez-Paz. In search of lost domain generalization. In *International*
524 *Conference on Learning Representations*, 2021.
- 525 [31] Christina Heinze-Deml, Jonas Peters, and Nicolai Meinshausen. Invariant causal prediction
526 for nonlinear models. *Journal of Causal Inference*, 6:20170016, 2018.
- 527 [32] Dan Hendrycks and Kevin Gimpel. A baseline for detecting misclassified and out-of-
528 distribution examples in neural networks. In *International Conference on Learning Rep-*
529 *resentations*, 2017.
- 530 [33] Fredrik D Johansson, David Sontag, and Rajesh Ranganath. Support and invertibility in domain-
531 invariant representations. In *The 22nd International Conference on Artificial Intelligence and*
532 *Statistics*, pages 527–536, 2019.
- 533 [34] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and
534 Tie-Yan Liu. Lightgbm: A highly efficient gradient boosting decision tree. *Advances in neural*
535 *information processing systems*, 30, 2017.
- 536 [35] IS Khalil, JC Doyle, and K Glover. *Robust and optimal control*, volume 2. Prentice hall New
537 York, 1996.
- 538 [36] Jakub Konečný, Brendan McMahan, and Daniel Ramage. Federated optimization: Distributed
539 optimization beyond the datacenter. *arXiv preprint arXiv:1511.03575*, 2015.
- 540 [37] Alexander Kraskov, Harald Stögbauer, and Peter Grassberger. Estimating mutual information.
541 *Physical Review E*, 2004.

- [38] David Krueger, Ethan Caballero, Joern-Henrik Jacobsen, Amy Zhang, Jonathan Binas, Dinghuai Zhang, Remi Le Priol, and Aaron Courville. Out-of-distribution generalization via risk extrapolation (rex). In *International conference on machine learning*, 2021.
- [39] Manabu Kuroki and Judea Pearl. Measurement bias and effect restoration in causal inference. *Biometrika*, 101, 2014.
- [40] Virgile Landeiro and Aron Culotta. Robust text classification under confounding shift. *Journal of Artificial Intelligence Research*, 63:391–419, 2018.
- [41] Kimin Lee, Kibok Lee, Honglak Lee, and Jinwoo Shin. A simple unified framework for detecting out-of-distribution samples and adversarial attacks. *Advances in neural information processing systems*, 31, 2018.
- [42] Daniel Levy, Yair Carmon, John C Duchi, and Aaron Sidford. Large-scale methods for distributionally robust optimization. *Advances in Neural Information Processing Systems*, 33: 8847–8860, 2020.
- [43] Haoliang Li, Sinno Jialin Pan, Shiqi Wang, and Alex C Kot. Domain generalization with adversarial feature learning. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 5400–5409, 2018.
- [44] Ying Li, Xingwei Wang, Rongfei Zeng, Praveen Kumar Donta, Ilir Murturi, Min Huang, and Schahram Dustdar. Federated domain generalization: A survey. *arXiv preprint arXiv:2306.01334*, 2023.
- [45] Xinting Liao, Weiming Liu, Pengyang Zhou, Fengyuan Yu, Jiahe Xu, Jun Wang, Wenjie Wang, Chaochao Chen, and Xiaolin Zheng. Foogd: Federated collaboration for both out-of-distribution generalization and detection. *Advances in Neural Information Processing Systems*, 37:132908–132945, 2024.
- [46] Yong Lin, Shengyu Zhu, Lu Tan, and Peng Cui. Zin: When and how to learn invariance without environment partition? *Advances in Neural Information Processing Systems*, 35: 24529–24542, 2022.
- [47] Zachary Lipton, Yu-Xiang Wang, and Alexander Smola. Detecting and correcting for label shift with black box predictors. In *International conference on machine learning*, pages 3122–3130. PMLR, 2018.
- [48] Roderick JA Little and Donald B Rubin. *Statistical analysis with missing data*. John Wiley & Sons, 2019.
- [49] Jiashuo Liu, Zheyang Shen, Yue He, Xingxuan Zhang, Renzhe Xu, Han Yu, and Peng Cui. Towards out-of-distribution generalization: A survey. *arXiv preprint arXiv:2108.13624*, 2021.
- [50] Jiashuo Liu, Tianyu Wang, Peng Cui, and Hongseok Namkoong. On the need for a language describing distribution shifts: Illustrations on tabular datasets. In *Thirty-seventh Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2023.
- [51] Mingsheng Long, Zhangjie Cao, Jianmin Wang, and Michael I Jordan. Conditional adversarial domain adaptation. *Advances in neural information processing systems*, 31, 2018.
- [52] Christos Louizos, Uri Shalit, Joris M Mooij, David Sontag, Richard Zemel, and Max Welling. Causal effect inference with deep latent-variable models. *Advances in neural information processing systems*, 30, 2017.
- [53] Fangrui Lv, Jian Liang, Shuang Li, Bin Zang, Chi Harold Liu, Ziteng Wang, and Di Liu. Causality inspired representation learning for domain generalization. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 8046–8056, June 2022.
- [54] Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Agüera y Arcas. Communication-efficient learning of deep networks from decentralized data. In *Artificial intelligence and statistics*, pages 1273–1282. PMLR, 2017.

- [55] Wang Miao, Zhi Geng, and Eric J Tchetgen Tchetgen. Identifying causal effects with proxy variables of an unmeasured confounder. *Biometrika*, 105:987–993, 2018.
- [56] Krikamol Muandet, David Balduzzi, and Bernhard Schölkopf. Domain generalization via invariant feature representation. In *Proceedings of the 30th International Conference on Machine Learning*, volume 28 of *Proceedings of Machine Learning Research*, pages 10–18, 2013.
- [57] Vivian Yvonne Nastl and Moritz Hardt. Do causal predictors generalize better to new domains? In *The Thirty-eighth Annual Conference on Neural Information Processing Systems*, 2024.
- [58] Judea Pearl. *Causality*. Cambridge university press, 2009.
- [59] Jonas Peters, Peter Bühlmann, and Nicolai Meinshausen. Causal inference by using invariant prediction: identification and confidence intervals. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 78:947–1012, 2016.
- [60] Steward TA Pickett. Space-for-time substitution as an alternative to long-term studies. In *Long-term studies in ecology: approaches and alternatives*, pages 110–135. Springer, 1989.
- [61] Sergei Popov, Stanislav Morozov, and Artem Babenko. Neural oblivious decision ensembles for deep learning on tabular data. *arXiv preprint arXiv:1909.06312*, 2019.
- [62] Parjanya Prajakta Prashant, Seyedeh Baharan Khatami, Bruno Ribeiro, and Babak Salimi. Scalable out-of-distribution robustness in the presence of unobserved confounders. In *The 28th International Conference on Artificial Intelligence and Statistics*, 2025.
- [63] Francesco Quinzan, Cecilia Casolo, Krikamol Muandet, Yucen Luo, and Niki Kilbertus. Learning counterfactually invariant predictors. *Transactions on Machine Learning Research*, 2024. ISSN 2835-8856.
- [64] Abbavaram Gowtham Reddy and Vineeth N Balasubramanian. Detecting and measuring confounding using causal mechanism shifts. *Advances in Neural Information Processing Systems*, 37:61677–61699, 2024.
- [65] Abbavaram Gowtham Reddy, Benin L Godfrey, and Vineeth N Balasubramanian. On causally disentangled representations. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 36, pages 8089–8097, 2022.
- [66] Mateo Rojas-Carulla, Bernhard Schölkopf, Richard Turner, and Jonas Peters. Invariant models for causal transfer learning. *Journal of Machine Learning Research*, 19:1–34, 2018.
- [67] Elan Rosenfeld, Pradeep Ravikumar, and Andrej Risteski. Domain-adjusted regression or: Erm may already learn features sufficient for out-of-distribution generalization. *arXiv preprint arXiv:2202.06856*, 2022.
- [68] Dominik Rothenhäusler, Nicolai Meinshausen, Peter Bühlmann, and Jonas Peters. Anchor regression: Heterogeneous data meet causality. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 83:215–246, 01 2021.
- [69] Shiori Sagawa, Pang Wei Koh, Tatsunori B Hashimoto, and Percy Liang. Distributionally robust neural networks for group shifts: On the importance of regularization for worst-case generalization. *arXiv preprint arXiv:1911.08731*, 2019.
- [70] Steffen Schneider, Evgenia Rusak, Luisa Eck, Oliver Bringmann, Wieland Brendel, and Matthias Bethge. Improving robustness against common corruptions by covariate shift adaptation. *Advances in neural information processing systems*, 33:11539–11551, 2020.
- [71] Bernhard Schölkopf. Causality for machine learning. In *Probabilistic and causal inference: The works of Judea Pearl*, pages 765–804, 2022.

- [72] Jessica Schrouff, Natalie Harris, Oluwasanmi O Koyejo, Ibrahim Alabdulmohsin, Eva Schnider, Krista Opsahl-Ong, Alexander Brown, Subhrajit Roy, Diana Mincu, Chrsitina Chen, Awa Dieng, Yuan Liu, Vivek Natarajan, Alan Karthikesalingam, Katherine A Heller, Silvia Chiappa, and Alexander D’Amour. Diagnosing failures of fairness transfer across distribution shift in real-world medical settings. In Alice H. Oh, Alekh Agarwal, Danielle Belgrave, and Kyunghyun Cho, editors, *Advances in Neural Information Processing Systems*, 2022.
- [73] E. H. Simpson. The interpretation of interaction in contingency tables. *Journal of the Royal Statistical Society: Series B (Methodological)*, 13(2):238–241, 2018.
- [74] Anurag Singh, Siu Lun Chau, Shahine Bouabid, and Krikamol Muandet. Domain generalisation via imprecise learning. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 45544–45570. PMLR, 2024.
- [75] Aman Sinha, Hongseok Namkoong, Riccardo Volpi, and John Duchi. Certifying some distributional robustness with principled adversarial training. *arXiv preprint arXiv:1710.10571*, 2017.
- [76] Gowthami Somepalli, Micah Goldblum, Avi Schwarzschild, C Bayan Bruss, and Tom Goldstein. Saint: Improved neural networks for tabular data via row attention and contrastive pre-training. *arXiv preprint arXiv:2106.01342*, 2021.
- [77] Greg Ver Steeg and Aram Galstyan. Information transfer in social media, 2011.
- [78] Greg Ver Steeg and Aram Galstyan. Information-theoretic measures of influence based on content dynamics, 2013.
- [79] Amos Storkey et al. When training and test sets are different: characterizing learning transfer. *Dataset shift in machine learning*, 30(3-28):6, 2009.
- [80] Masashi Sugiyama and Motoaki Kawanabe. *Machine learning in non-stationary environments: Introduction to covariate shift adaptation*. MIT press, 2012.
- [81] Baochen Sun and Kate Saenko. Deep coral: Correlation alignment for deep domain adaptation. In *Computer vision—ECCV 2016 workshops: Amsterdam, the Netherlands, October 8-10 and 15-16, 2016, proceedings, part III 14*, pages 443–450. Springer, 2016.
- [82] Baochen Sun and Kate Saenko. Deep coral: Correlation alignment for deep domain adaptation. In *Computer vision—ECCV 2016 workshops: Amsterdam, the Netherlands, October 8-10 and 15-16, 2016, proceedings, part III 14*, pages 443–450. Springer, 2016.
- [83] Remi Tachet des Combes, Han Zhao, Yu-Xiang Wang, and Geoffrey J Gordon. Domain adaptation with conditional distribution matching and generalized label shift. *Advances in Neural Information Processing Systems*, 33:19276–19289, 2020.
- [84] Jihoon Tack, Sangwoo Mo, Jongheon Jeong, and Jinwoo Shin. Csi: Novelty detection via contrastive learning on distributionally shifted instances. *Advances in neural information processing systems*, 33:11839–11852, 2020.
- [85] Katherine Tsai, Stephen R Pfohl, Olawale Salaudeen, Nicole Chiou, Matt Kusner, Alexander D’Amour, Sanmi Koyejo, and Arthur Gretton. Proxy methods for domain adaptation. In *International Conference on Artificial Intelligence and Statistics*, pages 3961–3969. PMLR, 2024.
- [86] Vladimir Vapnik. *The nature of statistical learning theory*. Springer science & business media, 1999.
- [87] Shanmukha Ramakrishna Vedantam, David Lopez-Paz, and David J. Schwab. An empirical investigation of domain generalization with empirical risk minimizers. In A. Beygelzimer, Y. Dauphin, P. Liang, and J. Wortman Vaughan, editors, *Advances in Neural Information Processing Systems*, 2021.

- [88] Jindong Wang, Cuiling Lan, Chang Liu, Yidong Ouyang, Tao Qin, Wang Lu, Yiqiang Chen, Wenjun Zeng, and S Yu Philip. Generalizing to unseen domains: A survey on domain generalization. *IEEE transactions on knowledge and data engineering*, 35:8052–8072, 2022.
- [89] Ruihan Wu, Chuan Guo, Yi Su, and Kilian Q Weinberger. Online adaptation to label distribution shift. *Advances in Neural Information Processing Systems*, 34:11340–11351, 2021.
- [90] Minghao Xu, Jian Zhang, Bingbing Ni, Teng Li, Chengjie Wang, Qi Tian, and Wenjun Zhang. Adversarial domain adaptation with domain mixup. In *Proceedings of the AAAI conference on artificial intelligence*, volume 34, pages 6502–6509, 2020.
- [91] Shen Yan, Huan Song, Nanxiang Li, Lincan Zou, and Liu Ren. Improve unsupervised domain adaptation with mixup training. *arXiv preprint arXiv:2001.00677*, 2020.
- [92] Jingkang Yang, Kaiyang Zhou, Yixuan Li, and Ziwei Liu. Generalized out-of-distribution detection: A survey. *International Journal of Computer Vision*, 132(12):5635–5662, 2024.
- [93] Haotian Ye, Chuanlong Xie, Tianle Cai, Ruichen Li, Zhenguo Li, and Liwei Wang. Towards a theoretical framework of out-of-distribution generalization. In A. Beygelzimer, Y. Dauphin, P. Liang, and J. Wortman Vaughan, editors, *Advances in Neural Information Processing Systems*, 2021.
- [94] Jingzhao Zhang, Aditya Menon, Andreas Veit, Srinadh Bhojanapalli, Sanjiv Kumar, and Suvrit Sra. Coping with label shift via distributionally robust optimisation. *arXiv preprint arXiv:2010.12230*, 2020.
- [95] Jingzhao Zhang, Aditya Krishna Menon, Andreas Veit, Srinadh Bhojanapalli, Sanjiv Kumar, and Suvrit Sra. Coping with label shift via distributionally robust optimisation. In *International Conference on Learning Representations*, 2021.
- [96] Kun Zhang, Bernhard Schölkopf, Krikamol Muandet, and Zhikun Wang. Domain adaptation under target and conditional shift. In *International conference on machine learning*, pages 819–827. Pmlr, 2013.
- [97] Liling Zhang, Xinyu Lei, Yichun Shi, Hongyu Huang, and Chao Chen. Federated learning with domain generalization. *arXiv preprint arXiv:2111.10487*, 2021.
- [98] Han Zhao, Remi Tachet Des Combes, Kun Zhang, and Geoffrey Gordon. On learning invariant representations for domain adaptation. In *International conference on machine learning*, pages 7523–7532, 2019.
- [99] Han Zhao, Chen Dan, Bryon Aragam, Tommi S Jaakkola, Geoffrey J Gordon, and Pradeep Ravikumar. Fundamental limits and tradeoffs in invariant representation learning. In *Journal of machine learning research*, volume 23, pages 1–49, 2022.
- [100] Kaiyang Zhou, Ziwei Liu, Yu Qiao, Tao Xiang, and Chen Change Loy. Domain generalization: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45:4396–4415, 2022.

717 Appendix

718 A Proofs of propositions

719 **Proposition 4.1.** For a covariate vector $\mathbf{X}$, label Y , with causal structure $\mathbf{X} \leftrightarrow Y$, i.e., some
 720 covariates cause Y and some covariates are caused by Y , environment variable E , a feature
 721 extractor ϕ , and prediction $\hat{Y}$, the predictive information $I(Y; \hat{Y})$ is decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativ.}} - \overbrace{\frac{I(\phi(\mathbf{X}); E|Y)}{2}}^{\text{Variation}} + \overbrace{\frac{I(Y; E)}{2}}^{\text{Label shift}} + \overbrace{\frac{I(\phi(\mathbf{X}); E)}{2}}^{\text{Feature shift}} - \overbrace{\frac{I(Y; E|\phi(\mathbf{X}))}{2}}^{\text{Concept shift}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (1)$$

722 where $I(\phi(\mathbf{X}); Y|\hat{Y})$ is the residual information in $\phi(\mathbf{X})$ for inferring Y that is not captured by the
 723 prediction $\hat{Y}$. The decomposition above also holds when $\phi(\mathbf{X})$ is replaced with $\mathbf{X}$.

724 *Proof.* From the causal graph shown in Figure 3 of the main paper, in the causal substructure:
 725 $\hat{Y} \leftarrow \phi(\mathbf{X}) \leftarrow \mathbf{X} \leftrightarrow Y$, $\hat{Y}$ provides no additional information in predicting Y given either $\phi(\mathbf{X})$ or
 726 $\mathbf{X}$. This is because, the path $\hat{Y} \leftarrow \phi(\mathbf{X}) \leftarrow \mathbf{X} \leftrightarrow Y$, between $\hat{Y}$ and Y is blocked by either $\{\phi(\mathbf{X})\}$
 727 or $\{\mathbf{X}\}$. This implies $\hat{Y} \perp\!\!\!\perp Y|\phi(\mathbf{X})$ and $\hat{Y} \perp\!\!\!\perp Y|\mathbf{X}$. Now consider the expansion of $I(Y; \hat{Y})$.

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; \phi(\mathbf{X}), \hat{Y}) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}); Y) + \underbrace{I(Y; \hat{Y}|\phi(\mathbf{X}))}_0 - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand and use } \hat{Y} \perp\!\!\!\perp Y | \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}), E; Y) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \mathbf{E}) \\ &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand}) \end{aligned} \quad (6)$$

728 Similarly,

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; \phi(\mathbf{X}), \hat{Y}) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \phi(\mathbf{X})) \\ &= I(\phi(\mathbf{X}); Y) + \underbrace{I(Y; \hat{Y}|\phi(\mathbf{X}))}_0 - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand and use } \hat{Y} \perp\!\!\!\perp Y | \phi(\mathbf{X})) \\ &= I(Y, E; \phi(\mathbf{X})) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Introduce } \mathbf{E}) \\ &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Expand}) \end{aligned} \quad (7)$$

729 Summing Equations (6) and (7) and grouping the terms gives the desired expression:

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); Y|E) - \frac{I(\phi(\mathbf{X}); E|Y)}{2} + \frac{I(Y; E)}{2} \\ &\quad + \frac{I(\phi(\mathbf{X}); E)}{2} - \frac{I(Y; E|\phi(\mathbf{X}))}{2} - I(\phi(\mathbf{X}); Y|\hat{Y}). \end{aligned}$$

730 To get the similar decomposition with $\mathbf{X}$ instead of $\phi(\mathbf{X})$, use $\hat{Y} \perp\!\!\!\perp Y | \mathbf{X}$ and introduce $\mathbf{X}$ into the
 731 expansion instead of $\phi(\mathbf{X})$. $\square$

732 **Proposition 4.2.** For a covariate vector $\mathbf{X}$, label Y , an environment variable E , a feature extractor
 733 ϕ , the prediction $\hat{Y}$, and an unobserved confounding variable U that cause both $\mathbf{X}$ and Y , and if
 734 either (i) $\mathbf{X} \rightarrow Y$ or (ii) $Y \rightarrow \mathbf{X}$, the predictive information $I(Y; \hat{Y})$ can be decomposed as follows:

$$I(Y; \hat{Y}) = \overbrace{I(\phi(\mathbf{X}); Y|E)}^{\text{Cond. informativeness}} - \overbrace{I(\phi(\mathbf{X}); Y|\hat{Y})}^{\text{Residual}} \quad (4)$$

735 *Proof.* Following the causal graph shown in Figure 3 of the main paper, we prove the proposition in
 736 two separate cases: $\mathbf{X} \rightarrow Y$ and $Y \rightarrow \mathbf{X}$.

737 **Case 1: $\mathbf{X} \rightarrow Y$.** In this case, we have the following inequality because conditioning on Y opens the
 738 path $\phi(\mathbf{X}) \leftarrow \mathbf{X} \rightarrow Y \leftarrow U \leftarrow E$ between $\phi(\mathbf{X})$ and E at the collider node Y :

$$I(\phi(\mathbf{X}); E|Y) \geq I(\phi(\mathbf{X}); E). \quad (8)$$

739 Additionally, considering the causal substructure: $Y \leftarrow \mathbf{X} \leftarrow U \leftrightarrow E$, we have the following
 740 inequality because conditioning on $\phi(\mathbf{X})$ partially blocks the information flow from Y to E at the
 741 node $\mathbf{X}$ as long as $\phi(\mathbf{X})$ encodes some information about $\mathbf{X}$:

$$I(Y; E) \geq I(Y; E|\phi(\mathbf{X})). \quad (9)$$

742 Now consider the following expansion from Equation (7) in the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (7)}) \\ &\leq I(\phi(\mathbf{X}); Y|E) + I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (8)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

743 Similarly, consider the following expansion from Equation (6) from the proof of Proposition 4.1:

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (6)}) \\ &\geq I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (9)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

744 That is, we have $I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \leq I(Y; \hat{Y}) \leq I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$.

745 **Case 2: $Y \rightarrow \mathbf{X}$.** In this scenario, we have the following inequality because conditioning on $\phi(\mathbf{X})$
 746 opens the path $E \rightarrow U \rightarrow \mathbf{X} \leftarrow Y$ between Y and E at the collider node $\mathbf{X}$ because $\phi(\mathbf{X})$ is a child
 747 of $\mathbf{X}$ [58].

$$I(Y; E|\phi(\mathbf{X})) \geq I(Y; E) \quad (10)$$

748 Additionally, we have the following inequality because conditioning on Y blocks the path $\phi(\mathbf{X}) \leftarrow$
 749 $\mathbf{X} \leftarrow Y \leftarrow U \leftrightarrow E$ at Y and hence less information flow between $\phi(\mathbf{X})$ and E .

$$I(\phi(\mathbf{X}); E) \geq I(\phi(\mathbf{X}); E|Y) \quad (11)$$

750 Now consider the following expansion from Equation (6) from the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(Y; E) + I(\phi(\mathbf{X}); Y|E) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (6)}) \\ &\leq I(\phi(\mathbf{X}); Y|E) + I(Y; E|\phi(\mathbf{X})) - I(Y; E|\phi(\mathbf{X})) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (10)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

751 Similarly, consider the following expansion from Equation (7) from the proof of Proposition 4.1.

$$\begin{aligned} I(Y; \hat{Y}) &= I(\phi(\mathbf{X}); E) + I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); E|Y) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Equation (7)}) \\ &\geq I(\phi(\mathbf{X}); Y|E) + I(\phi(\mathbf{X}); E) - I(\phi(\mathbf{X}); E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \quad (\text{Using (11)}) \\ &= I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \end{aligned}$$

752 That is, we have $I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y}) \leq I(Y; \hat{Y}) \leq I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$.

753 Since the upper bound and lower bounds are the same in both cases, we have the desired result
 754 $I(Y; \hat{Y}) = I(\phi(\mathbf{X}); Y|E) - I(\phi(\mathbf{X}); Y|\hat{Y})$. $\square$

755 **Proposition 4.3.** If $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and if $\mathbf{X}_I$ are informative variables such that
 756 $Y \not\perp \mathbf{X}_I \mid \mathbf{X}, E$, then we have the following inequalities:

$$\begin{aligned} (i) & I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E) \quad (ii) I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E) \\ (iii) & I(Y; E|\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\})) < I(Y; E|\phi_1(\mathbf{X})) \quad (iv) I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E \mid Y) > I(\phi_1(\mathbf{X}); E \mid Y) \end{aligned} \quad (5)$$

757 *Proof.* (i) By the chain rule of mutual information, we have: $I(\mathbf{X}, \mathbf{X}_I; Y|E) = I(\mathbf{X}; Y|E) +$
758 $I(\mathbf{X}_I; Y|E, \mathbf{X})$. We have $I(\mathbf{X}_I; Y|E, \mathbf{X}) > 0$ because $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid \mathbf{X}, E$. Hence, we have
759 $I(\mathbf{X}, \mathbf{X}_I; Y|E) > I(\mathbf{X}; Y|E)$. Since $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is
760 invariant to invertible transformations, we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); Y|E) > I(\phi_1(\mathbf{X}); Y|E)$.

761 (ii) We consider the causal graph shown in Figure 3. From the chain rule of mutual information,
762 we have: $I(\mathbf{X}, \mathbf{X}_I; E) = I(\mathbf{X}; E) + I(\mathbf{X}_I; E|\mathbf{X})$. Since $E \leftrightarrow U \rightarrow Y$, any covariate that is
763 informative to Y is also informative to U . Since $E \leftrightarrow U$, we have $I(\mathbf{X}_I; E|\mathbf{X}) > 0$, and hence
764 $I(\mathbf{X}, \mathbf{X}_I; E) > I(\mathbf{X}; E)$. Since $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is
765 invariant to invertible transformations, we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E) > I(\phi_1(\mathbf{X}); E)$.

766 (iii) Consider $I(Y; E, \mathbf{X}_I | \mathbf{X})$. It can be expressed in two ways:

$$I(Y; E, \mathbf{X}_I | \mathbf{X}) = I(Y; E | \mathbf{X}) + I(Y; \mathbf{X}_I | E, \mathbf{X}) = I(Y; \mathbf{X}_I | \mathbf{X}) + I(Y; E | \mathbf{X}_I, \mathbf{X})$$

767 Equating and rearranging the terms gives the following.

$$I(Y; E | \mathbf{X}) - I(Y; E | \mathbf{X}_I, \mathbf{X}) = I(Y; \mathbf{X}_I | \mathbf{X}) - I(Y; \mathbf{X}_I | E, \mathbf{X})$$

768 Since $Y \not\perp\!\!\!\perp \mathbf{X}_I \mid E, \mathbf{X}$, we have $I(Y; \mathbf{X}_I | \mathbf{X}) > I(Y; \mathbf{X}_I | E, \mathbf{X})$ because conditioning usually
769 reduces mutual information unless the additional conditioning variable opens any collider paths. Here
770 E is neither a collider nor a descendant of any collider. Thus, $I(Y; E | \mathbf{X}) > I(Y; E | \mathbf{X}_I, \mathbf{X})$. Since
771 $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is invariant to invertible transformations,
772 we have $I(Y; E | \phi_1(\mathbf{X})) > I(Y; E | \phi_2(\{\mathbf{X}_I \cup \mathbf{X}\}))$.

773 (iv) By the chain rule of mutual information, we have: $I(\mathbf{X}, \mathbf{X}_I; E | Y) = I(\mathbf{X}; E | Y) +$
774 $I(\mathbf{X}_I; E|\mathbf{X}, Y)$. Since $E \leftrightarrow U \rightarrow Y$, any covariate that is informative to Y is also informative to
775 U . Since $E \leftrightarrow U$, we have $I(\mathbf{X}_I; E|\mathbf{X}, Y) > 0$, and hence $I(\mathbf{X}, \mathbf{X}_I; E | Y) > I(\mathbf{X}; E | Y)$. Since
776 $\phi_1(\cdot), \phi_2(\cdot)$ are invertible functions and mutual information is invariant to invertible transformations,
777 we have $I(\phi_2(\{\mathbf{X} \cup \mathbf{X}_I\}); E | Y) > I(\phi_1(\mathbf{X}); E | Y)$. $\square$

778 B Experimental setup

779 In this section, we set out our experimental setup in detail. The impact of random seeds upon
780 the results is statistically insignificant across a great many settings [24]. For the results presented
781 in the main paper, we report the mean and standard deviation over five random hyperparameter
782 values—sampled from their respective domains in Table B1—for each dataset-method combination.
783 Inasmuch as models are evaluated on ID test and OOD test data, we estimate the mutual information
784 terms by recourse to the nonparametric entropy estimation toolbox [37, 77, 78]. Following [24],
785 we evaluate mutual information using both in-distribution (ID) test data, drawn from the training
786 domains, and out-of-distribution (OOD) test data, drawn from the test domains.

787 For the sake of consistency, we draw 20,000
788 random samples from each of the ID and OOD
789 test sets, although certain datasets contain fewer
790 than 20,000 samples. Throughout our experi-
791 ments, the sample size exerts a relatively negli-
792 gible effect upon the evaluated mutual informa-
793 tion terms. As Figure B1 illustrates, for instance,
794 the terms in Equation (1) computed for the repre-
795 sentations of the XGB model on the *income* dataset
796 remain relatively consistent as the sample size
797 is varied. Features $\phi(\mathbf{X})$ are extracted from the
798 layer immediately preceding the classification
799 head; for XGBoost, we combine the output margins with the SHAP values yielded by the model. All
800 experiments were carried out on a single NVIDIA RTX A6000 GPU.

801 In the experiments that study the impact of informative covariates upon predictive information, we
802 adhere to the categorization introduced by Nastl and Hardt [57], partitioning the covariates into
803 *causal*, *arguably causal*, and *all covariates*. Causal covariates are those whose influence upon
804 the target is clearly established, with well-attested directionality and negligible prospect of reverse
805 causation; strict reliance upon them, however, risks omitting relevant parents on account of gaps in
806 domain knowledge. Arguably causal covariates are those whose causal status is uncertain, satisfying

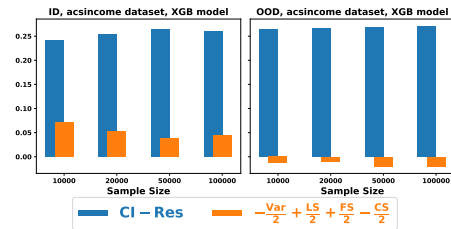


Figure B1: The evaluated terms of predictive information remain relatively consistent across a range of sample sizes on the *income* dataset.

Table B1: Hyperparameters, together with the candidate values considered in this paper.

Model	Hyperparameter	Search values
XGBoost	Learning rate	[0.1, 0.2, 0.3]
	Maximum tree depth	[4, 5, 6]
	Minimum child weight	[0.1, 1, 10]
	Gamma (min. loss reduction)	[0.0001, 0.001, 0.01]
	Subsample ratio	[0.5, 0.6, 0.7]
	Column subsample per tree	[0.5, 0.6, 0.7]
	L1 regularization (α)	[0.0001, 0.001, 0.01]
	L2 regularization (λ)	[0.0001, 0.001, 0.01]
MLP	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
	Number of epochs	[1, 2, 3]
GroupDRO	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Group-weights step size	[0.01, 0.05, 0.1]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
IRM	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	Dropout rate	[0.0, 0.1, 0.2]
	IRM penalty weight (λ)	[0.01, 0.05, 0.1]
	IRM penalty anneal iterations	[1, 2, 3]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]
VREX	Number of layers	[2, 3, 4]
	Hidden layer size	[256, 512, 1024]
	VREX penalty anneal iterations	[1, 2, 3]
	VREX penalty weight (λ)	[0.1, 10, 100]
	Dropout rate	[0.0, 0.1, 0.2]
	Learning rate	[0.01, 0.02, 0.05]
	Weight decay (L2)	[0.0001, 0.001, 0.01]
	Batch size	[4096]

807 at least one of the following conditions: (1) being known to be causal, (2) exerting a plausible but
 808 potentially bidirectional influence, or (3) bearing a likely though unconfirmed causal effect. These
 809 groupings approximate the true causal parents insofar as available knowledge permits, notwithstanding
 810 that the relationships may be confounded. Certain datasets additionally contain *anti-causal*
 811 covariates, upon which the target likely acts without the converse holding. The *all covariates* set,
 812 finally, comprehends every observed variable irrespective of causal status. The sign consistency
 813 metric introduced in the main paper § 5 is defined as follows. For each $(m, \sigma_m) \in M$ where
 814 $M = \{(\text{conditional informativeness}, +1), (\text{variation}, -1), (\text{label shift}, +1), (\text{feature shift}, +1),$

815 $(\text{concept shift}, -1)$, and $(\text{residual}, -1)$, for a specific dataset, we compute:

$$C_m = \frac{1}{2|P|} \sum_{(s,t) \in P} [\mathbb{I}(\sigma_m \cdot (\mu_t^{\text{tr}}(m) - \mu_s^{\text{tr}}(m)) > 0) + \mathbb{I}(\sigma_m \cdot (\mu_t^{\text{te}}(m) - \mu_s^{\text{te}}(m)) > 0)]$$

816 Here P denotes the set of covariate-set pairs (s, t) with $t \supset s$; s , for example, might be the causal
817 covariates and t the arguably causal covariates. The factor of 2 in $2P$ accounts for both the train and
818 the test data, while $\mu_s^{\text{tr/te}}(m)$ denotes the mean of measure m for covariate set s in the training/testing
819 data, and $\mathbb{I}(\cdot)$ the indicator function; here $|P| = 3$ for the covariate settings $\{\text{causal}, \text{arguably causal},$
820 $\text{all}\}$. To arrive at the final sign consistency metric value for each m , we average across datasets:
821 $\bar{C}_m = \frac{1}{|D|} \sum_{d \in D} C_m^{(d)}$, where $|D|$ is the number of datasets.

822 C Additional results on real-world datasets

823 In this section, we present additional results on real-world datasets. Figures C2 and C3 exhibit, for
824 each dataset, the comparison of the terms *conditional informativeness – residual* and *–variation / 2 +*
825 *label shift / 2 + feature shift / 2 – concept shift / 2*, whose aggregation appears in the main paper as
826 Figure 4 (Left). The salient takeaway from these results is that, per Proposition 4.1, the sum of the
827 blue and orange terms is positively correlated with overall model performance. At the same time,
828 owing to the latent hidden confounding shift in real-world data, the contribution of the difference
829 *conditional informativeness – residual* toward the overall predictive information is the greater when
830 set against that of *–variation / 2 + label shift / 2 + feature shift / 2 – concept shift / 2*. This furnishes a
831 motivation for maximizing conditional informativeness in the interest of superior OOD generalization
832 under a hidden confounding shift.

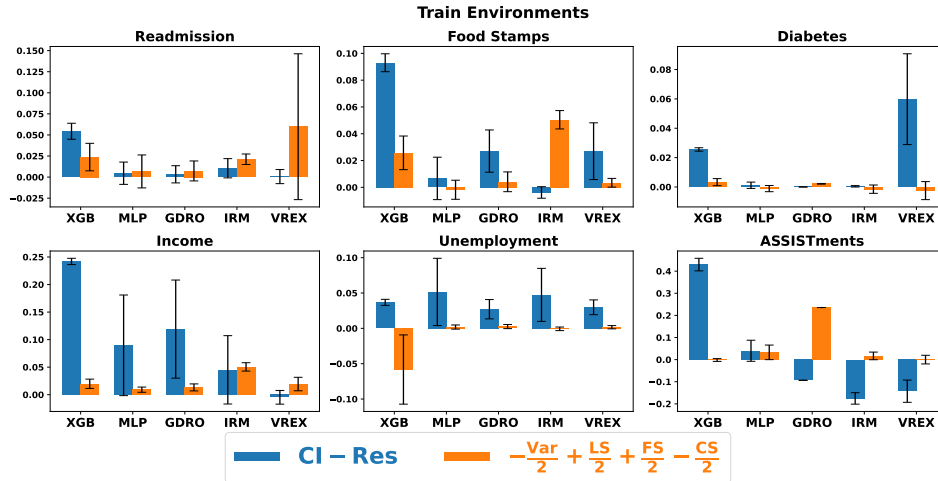


Figure C2: Dataset-specific results on the ID data.

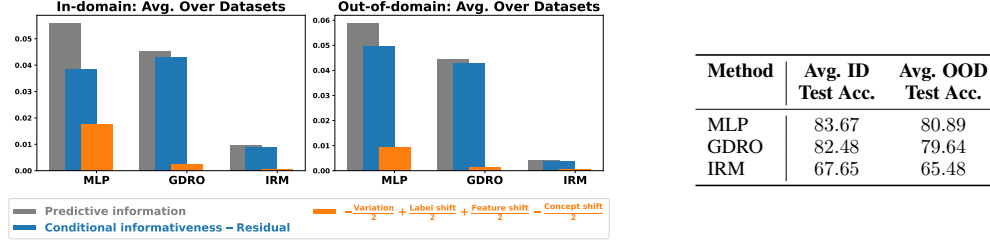


Figure C4: The difference *conditional informativeness* – *residual* depicted in the plots is positively correlated with the average ID and OOD test accuracy over the five datasets tabulated to the right. The contribution of the difference *conditional informativeness* – *residual* toward the overall predictive information is the greater when set against that of $-\text{variation} / 2 + \text{label shift} / 2 + \text{feature shift} / 2 - \text{concept shift} / 2$.

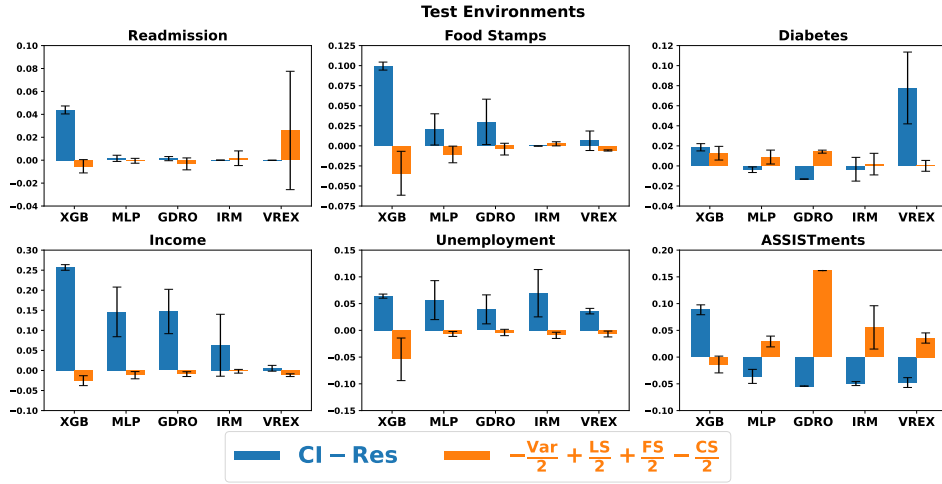


Figure C3: Dataset-specific results on the OOD data.

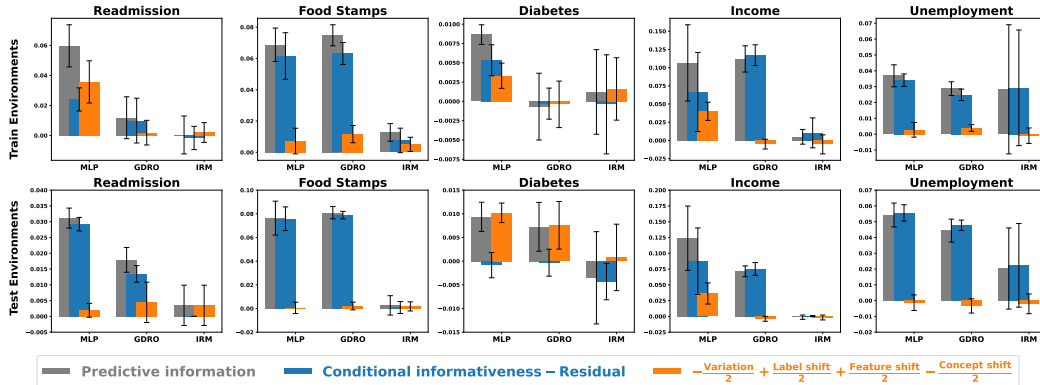


Figure C5: Dataset-specific results: a comparison of the terms constituting the predictive information.

Table C2: A comparison of the ID and OOD test accuracies attained by the models.

Method	Readmission		Food stamps		Diabetes	
	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.
MLP	65.78 ± 0.14	61.75 ± 0.22	84.74 ± 0.03	82.00 ± 0.06	87.66 ± 0.02	83.22 ± 0.05
GDRO	60.47 ± 1.08	57.60 ± 0.42	84.47 ± 0.04	81.41 ± 0.09	87.47 ± 0.10	82.90 ± 0.10
IRM	50.28 ± 7.35	50.85 ± 1.74	80.91 ± 0.00	78.01 ± 0.00	42.54 ± 36.5	43.48 ± 31.9

	Income		Unemployment	
	ID test acc.	OOD test acc.	ID test acc.	OOD test acc.
MLP	82.92 ± 0.05	81.46 ± 0.01	97.27 ± 0.02	96.04 ± 0.05
GDRO	82.70 ± 0.05	80.16 ± 0.45	97.28 ± 0.01	96.10 ± 0.00
IRM	67.91 ± 0.00	60.20 ± 0.00	96.61 ± 0.00	94.84 ± 0.00

Figures C6–C15 further exhibit, for each method and dataset, the comparison of the various terms of the information decomposition in (1). From these results we observe that the values plotted in the second column (*variation*) and the third column (*feature shift*) are of comparable magnitude, so that, by Equation 1, their difference lies close to zero and contributes but negligibly to the overall predictive decomposition. In like manner, the values plotted in the fourth column (*label shift*) and the fifth column (*concept shift*) are of comparable magnitude, so that, by Equation 1, their difference again approaches zero and contributes but negligibly to the overall predictive decomposition. In nearly every setting, accordingly, the difference *conditional informativeness* - *residual* constitutes the preponderant contributor to the predictive information.

Hyperparameter tuning: We additionally perform hyperparameter tuning over the model parameters. We conduct a grid search over hyperparameter values and select the optimal configuration on the basis of ID validation accuracy, rather than OOD validation accuracy. In all, we run 1500 experiments (5 datasets, 3 models, and 100 hyperparameter choices sampled from a grid of hyperparameter values); to obtain the mean and standard deviation of the accuracies and mutual information terms, we execute each experiment across 5 random seeds, notwithstanding that the effect of random seeds is in many cases insignificant. The results are displayed in Figure C4. By Proposition 4.1, the sum of the *blue* and *orange* terms is positively correlated with overall model performance; and, once again, owing to the latent hidden confounding shift in real-world data, the contribution of the difference *conditional informativeness* - *residual* toward the overall predictive information is the greater when set against that of $-variation / 2 + label\ shift / 2 + feature\ shift / 2 - concept\ shift / 2$. The dataset-specific results reported in Figure C5 and Table C2 evince analogous trends.

Potential limitations with predictive information decomposition analysis: Although the predictive information $I(Y; \hat{Y})$ measures the statistical dependence between predictions and labels, it affords no guarantee of accuracy. A binary classifier that systematically predicts the *wrong* label (e.g., $\hat{Y} = 1$ when $Y = 0$), for instance, attains the maximal $I(Y; \hat{Y}) = 1$ bit—for binary variables—by dint of perfect anti-correlation, and yet yields 0% accuracy. This apparent paradox arises because mutual information quantifies the *reduction of uncertainty* rather than correctness per se. The remaining decomposition terms afford further insight: conditional informativeness $I(\phi(X); Y|E)$ may cloak a deleterious variation $I(\phi(X); E|Y)$, while feature shift $I(\phi(X); E)$ can inflate $I(Y; \hat{Y})$ by way of spurious correlations. Even so, the decomposition retains its value for the analysis of reasonably performant models, as our experiments attest—it discloses whether the predictive information issues from generalizable patterns or from environment-specific artifacts, thereby enabling targeted improvements while preserving interpretability.

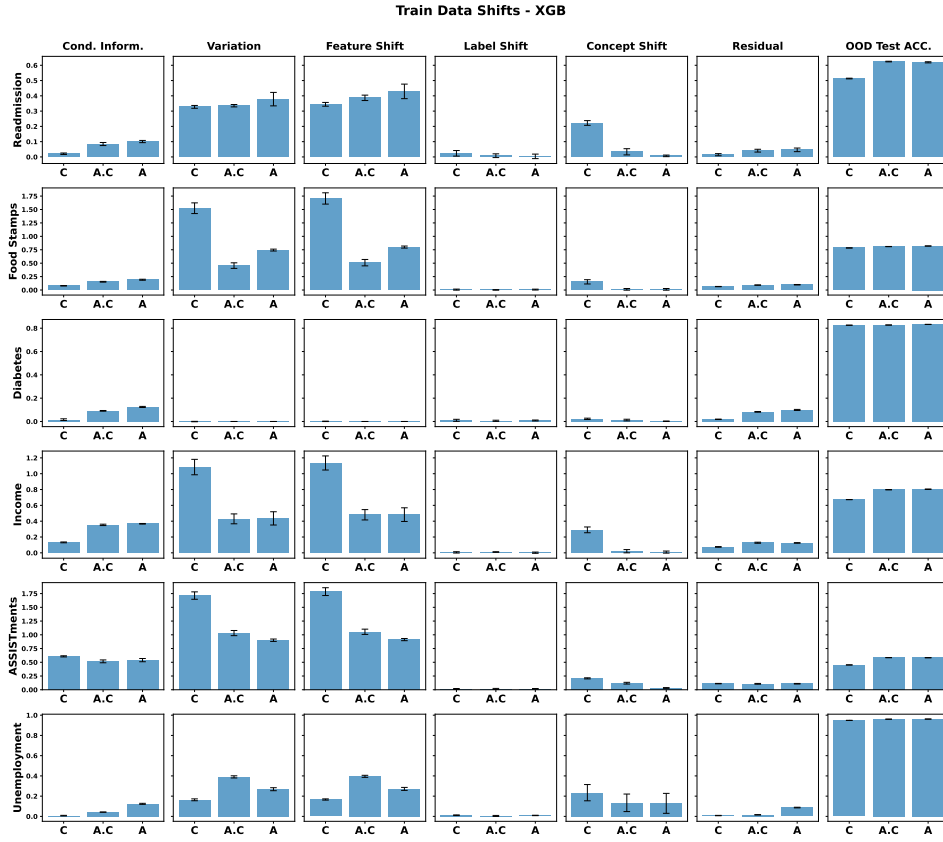


Figure C6: Decomposition of the information metrics on the training data for XGBoost, across the feature groups: (C) causal, (A.C) arguably causal, (A) all.

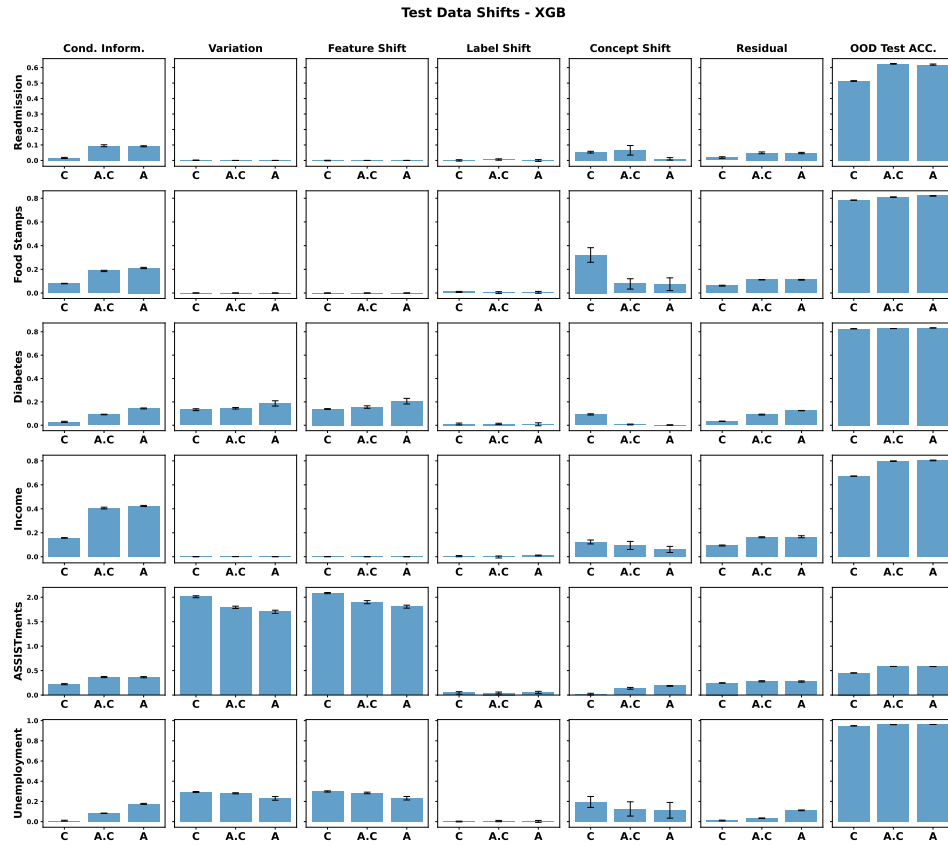


Figure C7: Decomposition of the information metrics on the test data for XGBoost, across the feature groups: (C) causal, (A.C) arguably causal, (A) all.

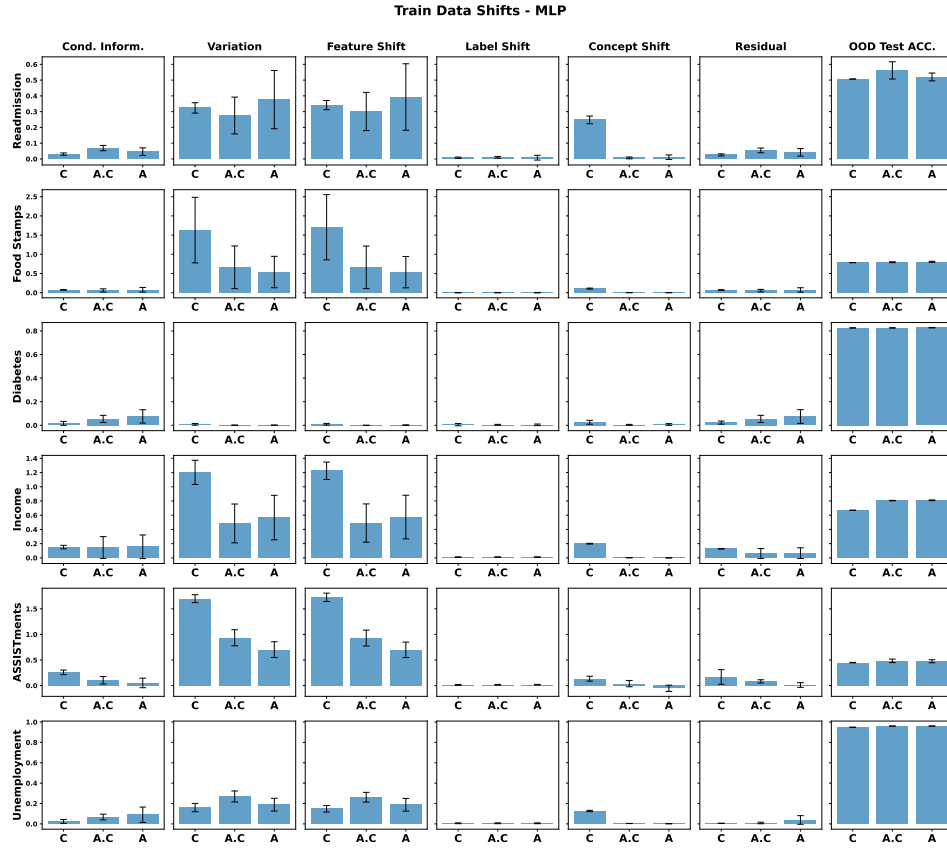


Figure C8: Decomposition of the information metrics on the training data for the MLP, across the feature groups: (C) causal, (A.C) arguably causal, (A) all.

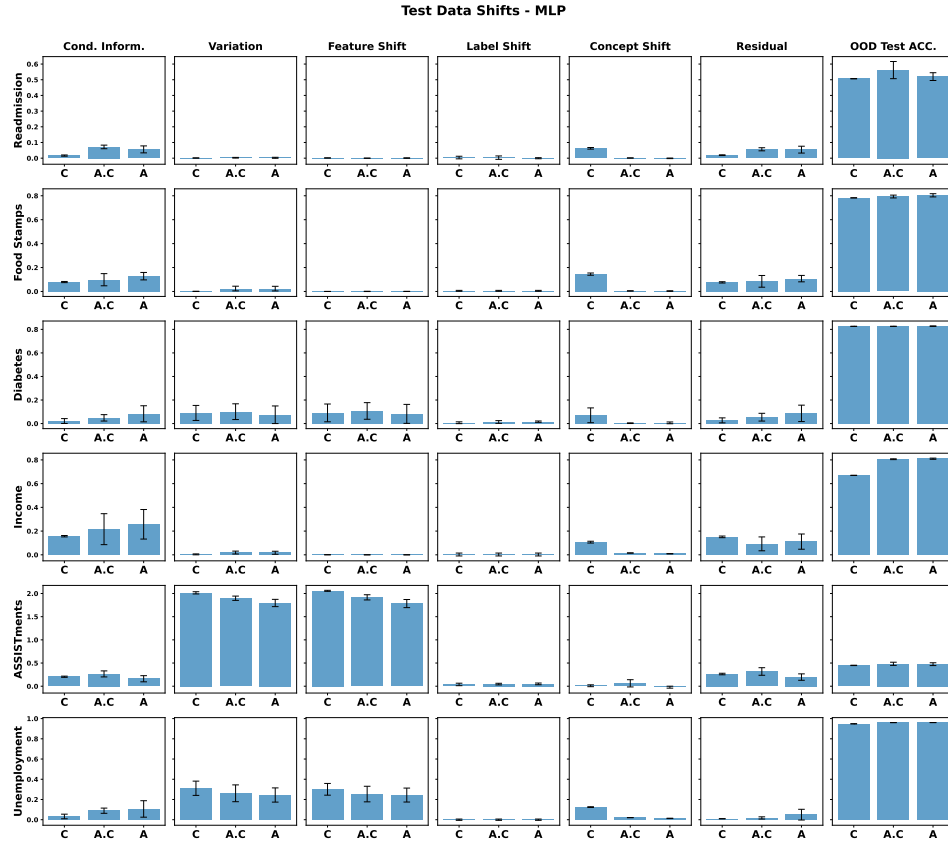


Figure C9: Decomposition of the information metrics on the test data for the MLP, across the feature groups: (C) causal, (A.C) arguably causal, (A) all.

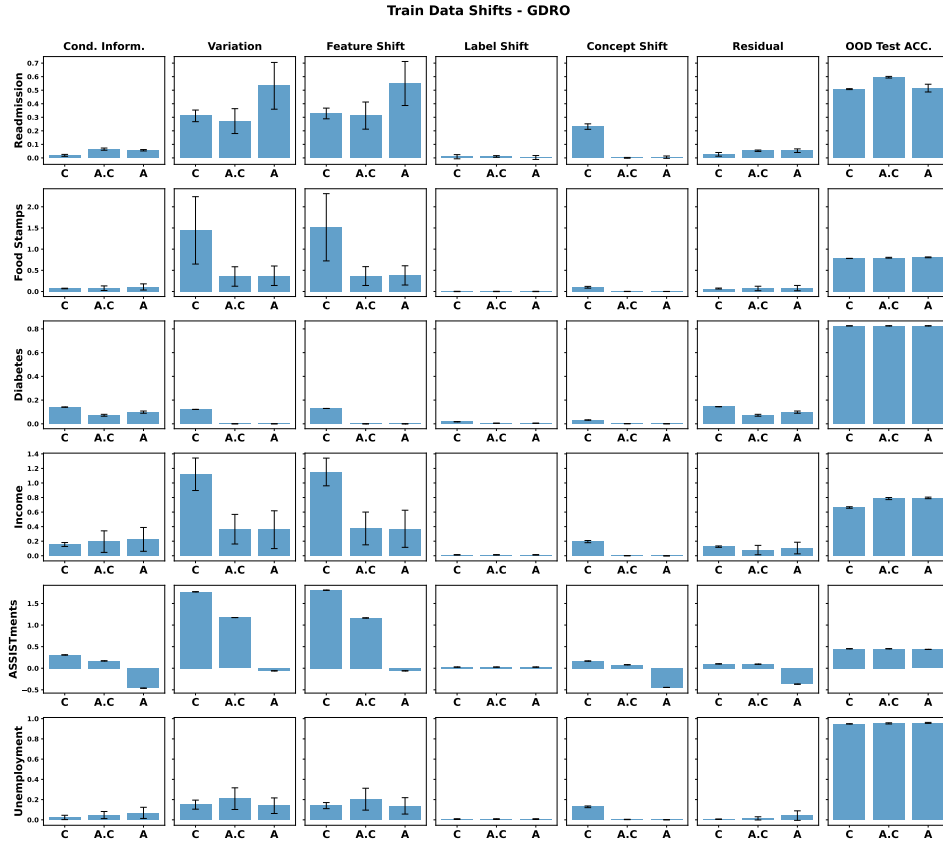


Figure C10: Decomposition of the information metrics on the training data for GDRO, across the feature groups: (C) causal, (A.C) arguably causal, (A) all.

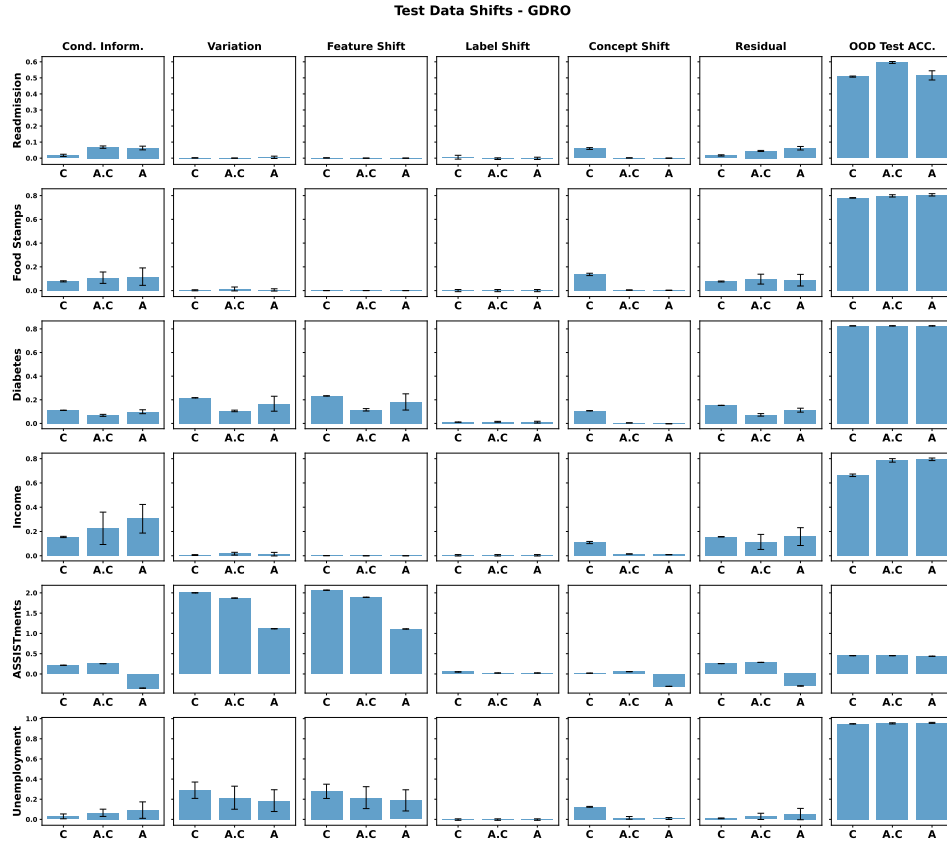


Figure C11: Decomposition of the information metrics on the test data for GDRO, across the feature groups: (C) causal, (A.C) arguably causal, (A) all.

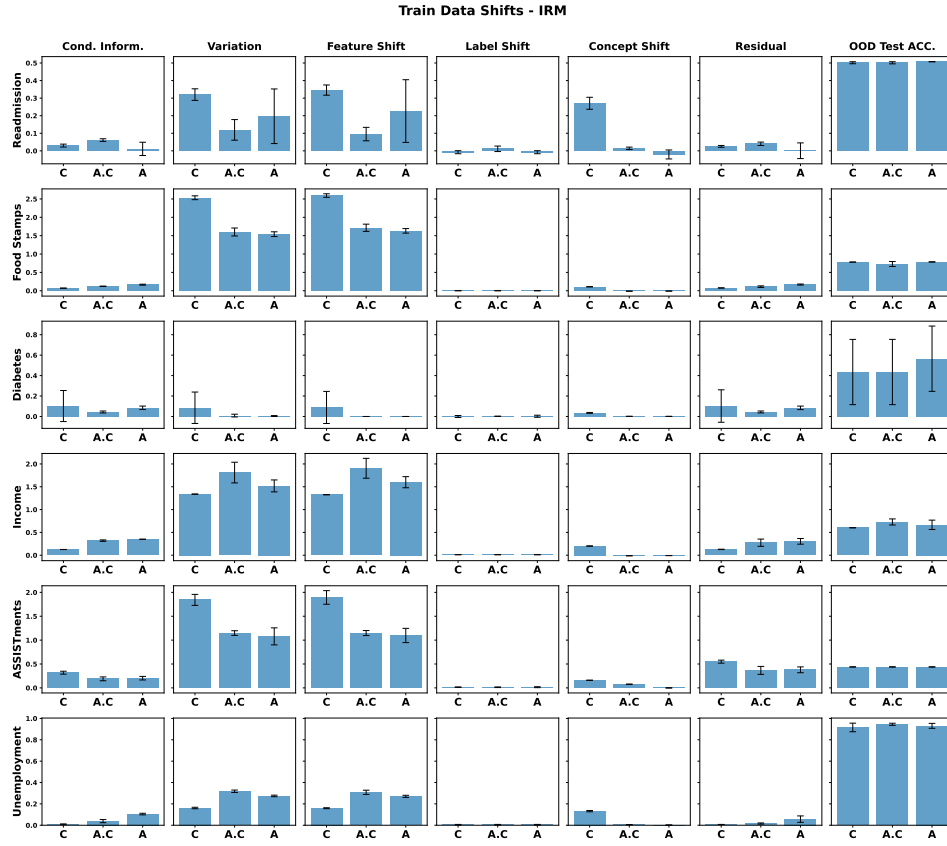


Figure C12: Decomposition of the information metrics on the training data for IRM, across the feature groups: (C) causal, (A.C) arguably causal, (A) all.

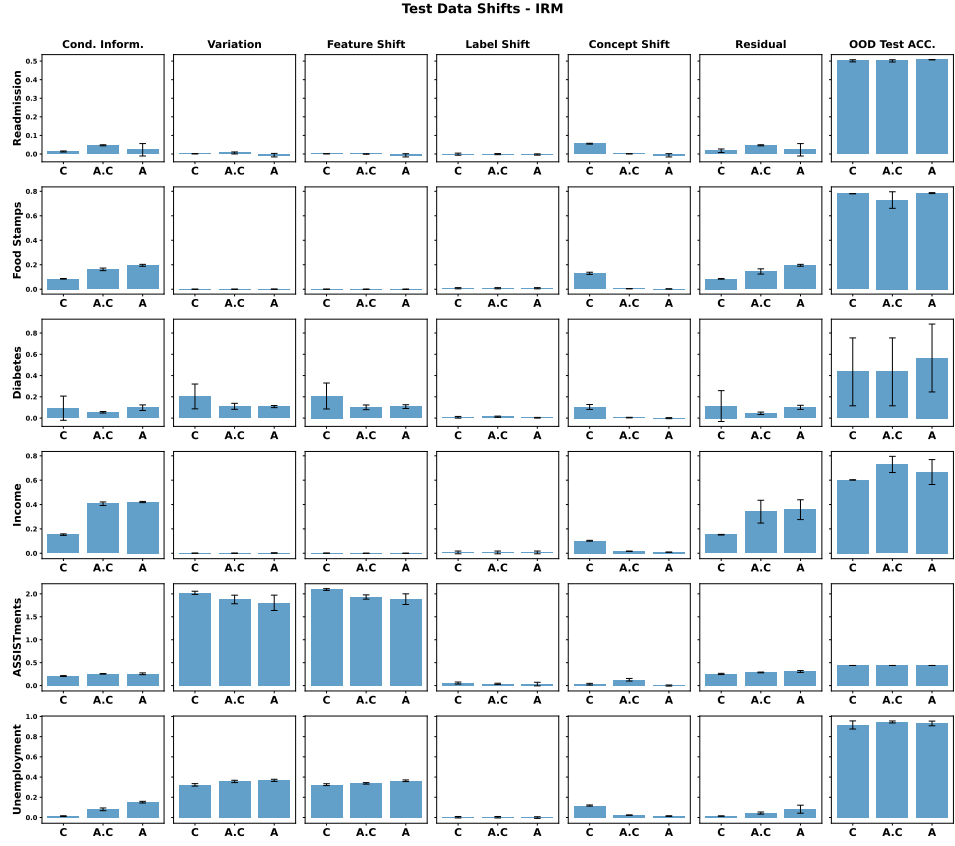


Figure C13: Decomposition of the information metrics on the test data for IRM, across the feature groups: (C) causal, (A.C) arguably causal, (A) all.

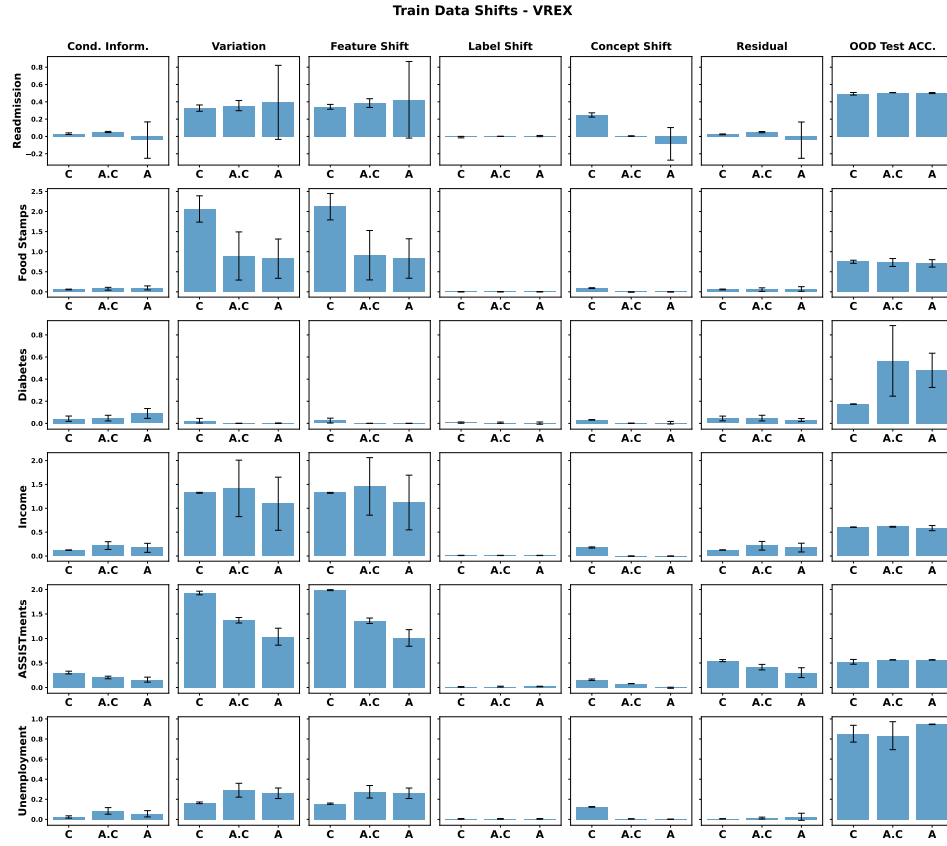


Figure C14: Decomposition of the information metrics on the training data for VREX, across the feature groups: (C) causal, (A.C) arguably causal, (A) all.

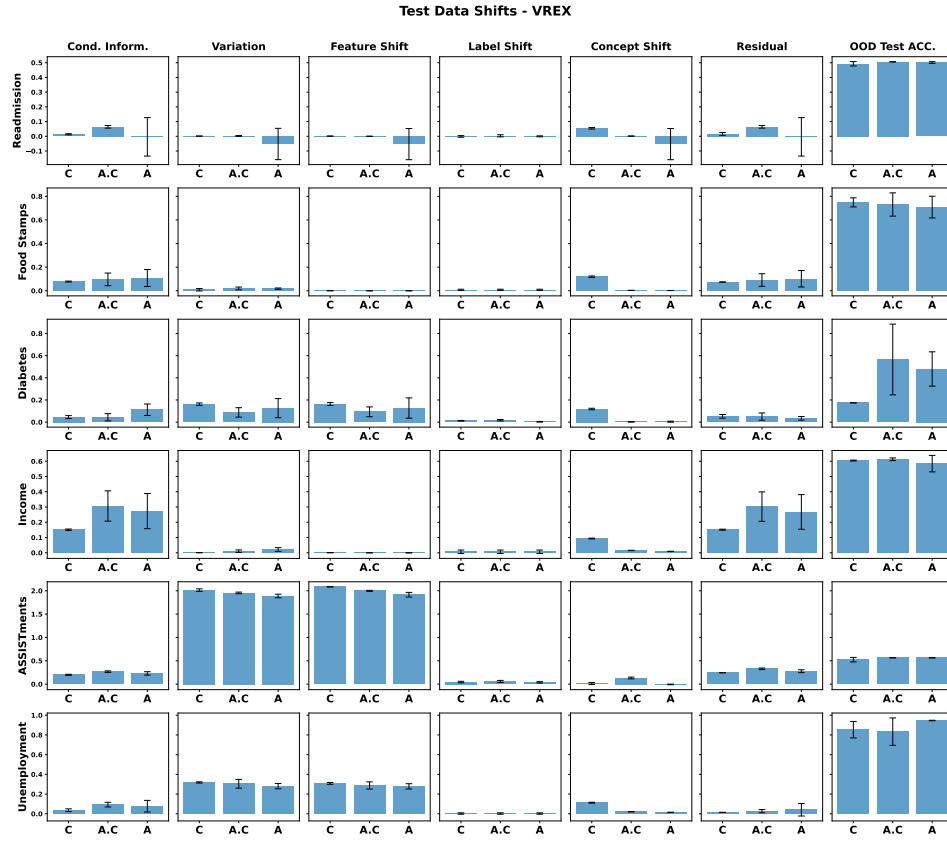


Figure C15: Decomposition of the information metrics on the test data for VREX, across the feature groups: (C) causal, (A.C) arguably causal, (A) all.

D Results on synthetic datasets

We conduct a series of experiments on synthetic datasets in order to investigate: (i) the impact of differing informativeness criteria, (ii) the manner in which the degree of confounder overlap across environments bears upon model performance, and (iii) the influence exerted by the number of causal as against anti-causal variables upon performance. The data-generating process conforms to the structural equations in (12), with the causal structure $U \rightarrow X, U \rightarrow Y, U \rightarrow \mathbf{X}_I$, and $X \rightarrow Y$, wherein U is an unobserved confounder, X an observed covariate, Y the target variable, and $\mathbf{X}_I$ the additional informative covariates.

$$\begin{aligned} U &\sim \mathcal{N}(\mu_u^e, \sigma_u) & X &\leftarrow f_X(U) + \mathcal{N}(\mu_x, \sigma_x) \\ X_i &\leftarrow f_i(U) + \mathcal{N}(\mu_i, \sigma_i); X_i \in \mathbf{X}_I & Y &\leftarrow f_Y(X, U) + \mathcal{N}(\mu_y, \sigma_y) \end{aligned} \quad (12)$$

We explore a variety of functional forms for f_X, f_i , and f_Y , whose corresponding results are set out in our analysis. Domain shifts are induced by systematically varying the environment-specific mean parameter μ_u^e of the confounder distribution, an expedient that permits us to examine how differing degrees of distributional shift affect model performance across environments.

Informativeness vs. accuracy: In this set of experiments, we begin with the linear structural equations $f_X = 0.3U$, $f_i = 0.1U$, and $f_Y = X - 2U$, adopting $\mu_u^e \in \{-2, 2\}$ for the training environments and $\mu_u^e \in \{0, 4\}$ for the test environments. For the results presented in Figure 5 of the main paper, we consider a single hidden confounding variable together with $|\mathbf{X}_I| = 20$ informative covariates. As the number of informative covariates is increased from 0 to 20, we observe (i) a reduction in mean squared error, (ii) an improvement in conditional informativeness, (iii) an enhancement of feature shift, and (iv) a diminution of concept shift. We extend these findings by examining two further informativeness criteria. We first analyze the case of one hidden confounder and a single informative covariate ($|\mathbf{X}_I| = 1$) across the varying noise levels $\sigma_i \in \{0, 0.1, \dots, 2.0\}$; Figure D16 demonstrates that, as the noise decreases from 2.0 to 0.0, we observe (i) a reduced mean squared error, (ii) improved conditional informativeness, (iii) enhanced feature shift, and (iv) diminished concept shift, the variation term remaining zero throughout, in consonance with the other synthetic experiments. We finally investigate a setting endowed with 20 hidden confounders and a corresponding 20 informative covariates, using the modified structural equations $f_X = 0.3U$, $f_i = 0.2U$, and $f_Y = X - 2U$; Figure D17 shows that, as informative covariates corresponding to the hidden confounders are introduced, we again observe (i) a reduced mean squared error, (ii) improved conditional informativeness, (iii) enhanced feature shift, and (iv) diminished concept shift. Taken together, these experiments demonstrate how differing informativeness criteria conduce to the attainment of OOD generalization.

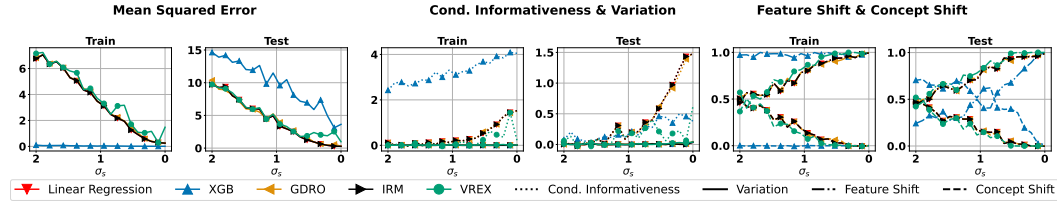


Figure D16: The introduction of a low-noise proxy variable conduces to a reduction in MSE and to increases in conditional informativeness and feature shift, while diminishing concept shift.

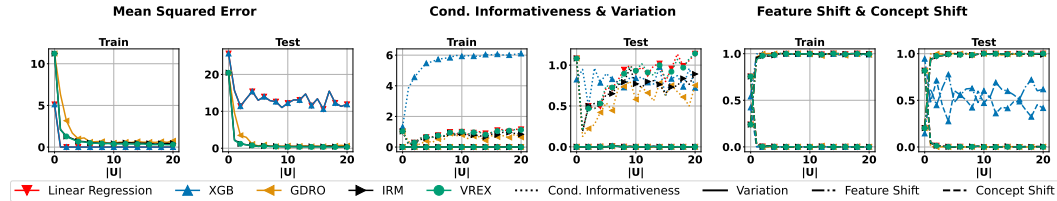


Figure D17: The introduction of additional proxy variables $\mathbf{X}_I$ of a set of hidden confounding variables $\mathbf{U}$ that are informative of Y conduces to a reduction in MSE and to increases in conditional informativeness and feature shift, while diminishing concept shift.

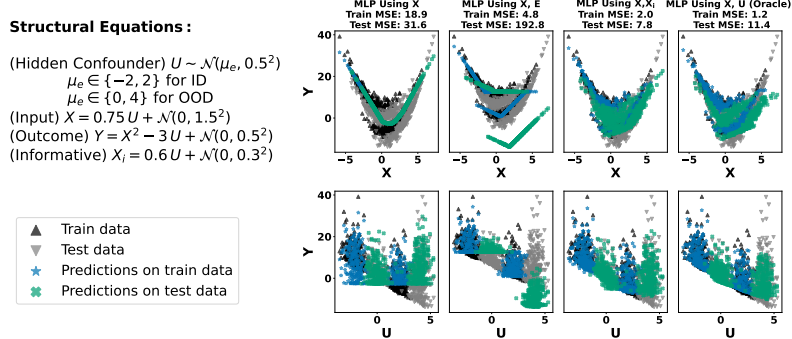


Figure D18: The performance of the MLP in the low-overlap non-linear setting, across the feature sets: (i) X ; (ii) X together with E (the environment statistics); (iii) X together with X_i ; and (iv) X together with U (the oracle).

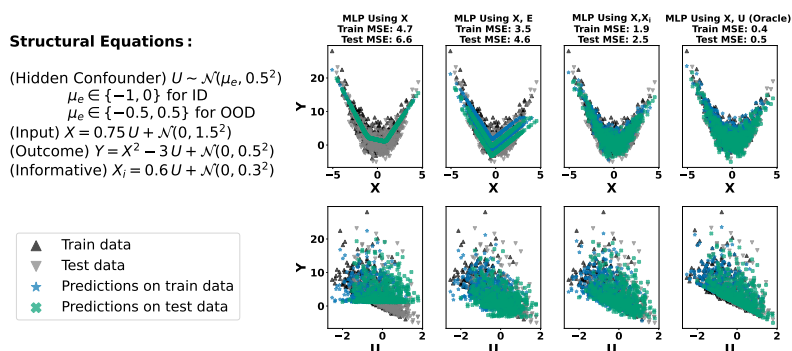


Figure D19: The performance of the MLP in the high-overlap non-linear setting, across the feature sets: (i) X ; (ii) X together with E (the environment statistics); (iii) X together with X_i ; and (iv) X together with U (the oracle).

Overlapping confounder support vs. accuracy: As established in the main paper, recent work by Prashant et al. [62] advanced an OOD generalization method for hidden confounding shift predicated on the assumption that the test confounder support remains contained within the training support. Whereas our results demonstrated the successful learning of correct relationships in synthetic linear data whenever the observed covariates harbor sufficient information about the hidden confounders, we now extend the analysis to nonlinear data, employing MLP and XGB models under both low and high confounding support conditions (Figures D18–D21). In the low-overlap settings—characterized by distant μ_e between the ID and OOD data—models relying upon X alone learn an inadequate global function that fails to discriminate among environments. Performance improves once environment-specific summary statistics of the observed covariates are incorporated, since these assist in capturing environment-specific relationships; the most substantial gains, however, materialize only upon the exploitation of additional informative covariates, a fact that underscores their importance. For completeness, we furnish both the oracle model results and the U - Y relationship scatter plots, so as to elucidate the learned input–output mappings. These experiments confirm, as anticipated, superior performance under high confounding overlap; more consequentially, they reveal promising results in the low-overlap regime—a hitherto unstudied scenario that calls into question the common-support assumption prevalent in the literature.

Number of causal vs anti-causal variables: Recent benchmarks [24] establish that baseline methods (e.g., XGB, MLP) consistently match or surpass dedicated OOD generalization techniques (e.g., GDRO, IRM, VREX) on real-world datasets—a phenomenon coincident with the prevalence of the causal structure $\mathbf{X} \rightarrow Y$ in tabular data. To adjudicate whether the ascendancy of the baselines is attributable to the causal structure $\mathbf{X} \rightarrow Y$, we vary the causal-to-anti-causal ratio $\rho = |\mathbf{X}_C|/|\mathbf{X}_A| \in \{0.0, \dots, 1.0\}$ with $|\mathbf{X}| = 50$ and $\mathbf{X} = \mathbf{X}_C \cup \mathbf{X}_A$ (so that $\rho = 0.5$, for instance, yields $|\mathbf{X}_C| = |\mathbf{X}_A| = 25$). The results (Figure D22) attest to the resilience of the baselines even under the causal

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-2, 2\}$ for ID
 $\mu_e \in \{0, 4\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X^2 - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

▲ Train data
▼ Test data
★ Predictions on train data
✱ Predictions on test data

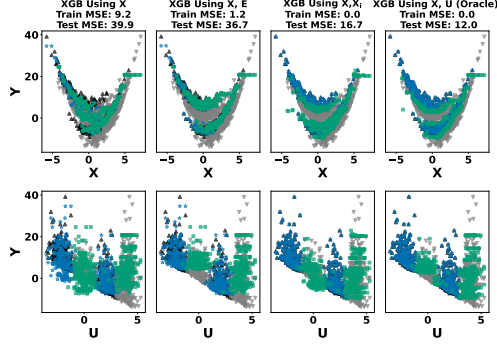


Figure D20: The performance of XGB in the low-overlap non-linear setting, across the feature sets: (i) X ; (ii) X together with E (the environment statistics); (iii) X together with X_i ; and (iv) X together with U (the oracle).

Structural Equations :

(Hidden Confounder) $U \sim \mathcal{N}(\mu_e, 0.5^2)$
 $\mu_e \in \{-1, 0\}$ for ID
 $\mu_e \in \{-0.5, 0.5\}$ for OOD
(Input) $X = 0.75U + \mathcal{N}(0, 1.5^2)$
(Outcome) $Y = X^2 - 3U + \mathcal{N}(0, 0.5^2)$
(Informative) $X_i = 0.6U + \mathcal{N}(0, 0.3^2)$

▲ Train data
▼ Test data
★ Predictions on train data
✱ Predictions on test data

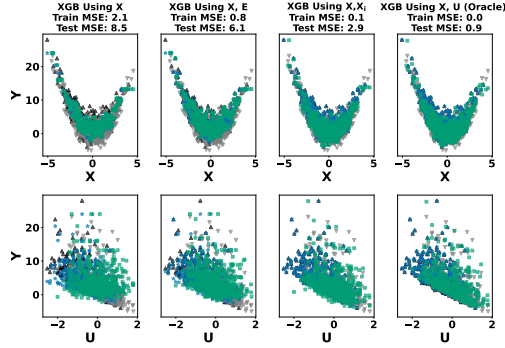


Figure D21: The performance of XGB in the low-overlap non-linear setting, across the feature sets: (i) X ; (ii) X together with E (the environment statistics); (iii) X together with X_i ; and (iv) X together with U (the oracle).

921 structure $\mathbf{X}_C \rightarrow Y \rightarrow \mathbf{X}_A$, albeit no single method dominates universally (Figures D23, D24); the
922 baselines attain superior PMA-OOD scores [24] (the fraction of maximum OOD accuracy). We
923 evaluate: (1) ERM-based methods (XGBoost [15], LightGBM [34], MLP, ResNet [27], SAINT [76],
924 TabTransformer, NODE [61], FT-Transformer, ExpGrad [2]); (2) OOD generalization methods
925 (IRM [6], IB-IRM, IB-ERM [30], CausIRL [17], DANN [3], MMD [43], DeepCORAL [81], V-
926 REx [38], Domain Mixup [90, 91]); and (3) robust optimization methods (DRO [42], GroupDRO [69],
927 Label DRO, Adversarial Label DRO [94]).

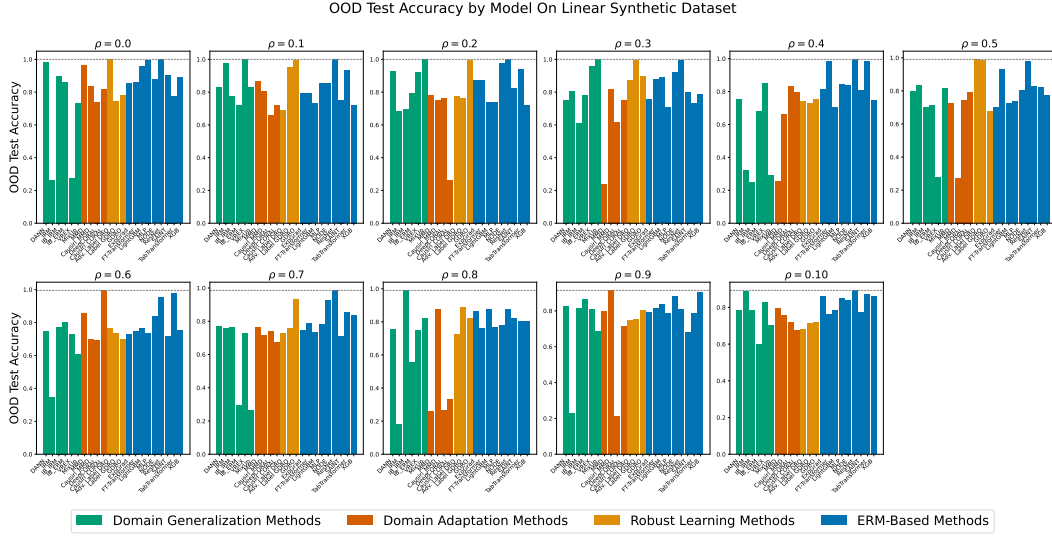


Figure D23: OOD test accuracy in the linear setting, reported per model across differing values of ρ .

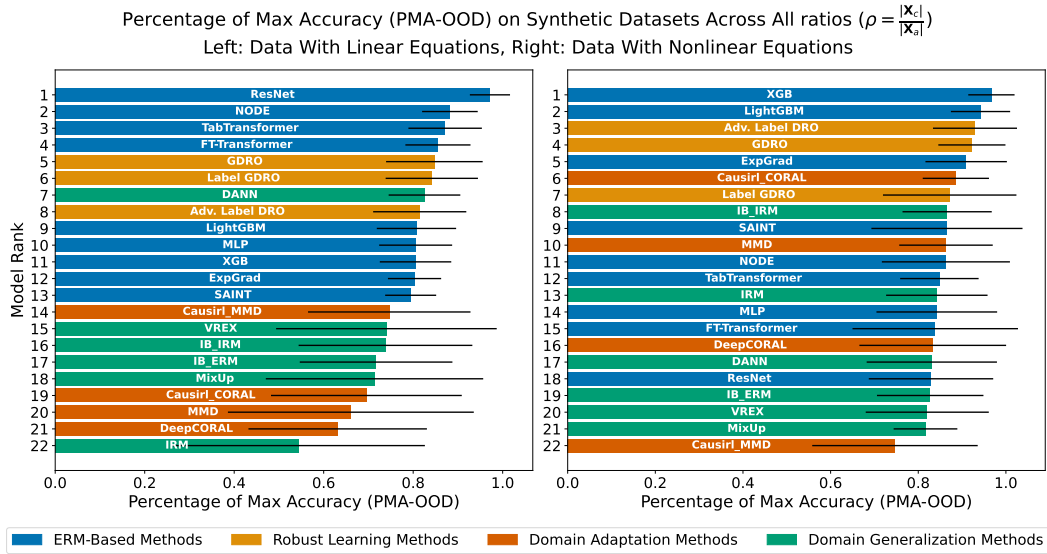


Figure D22: ERM-based methods attain a superior percentage of maximum OOD (PMA-OD) test accuracy on the synthetic datasets, in keeping with the real-world results of [24].

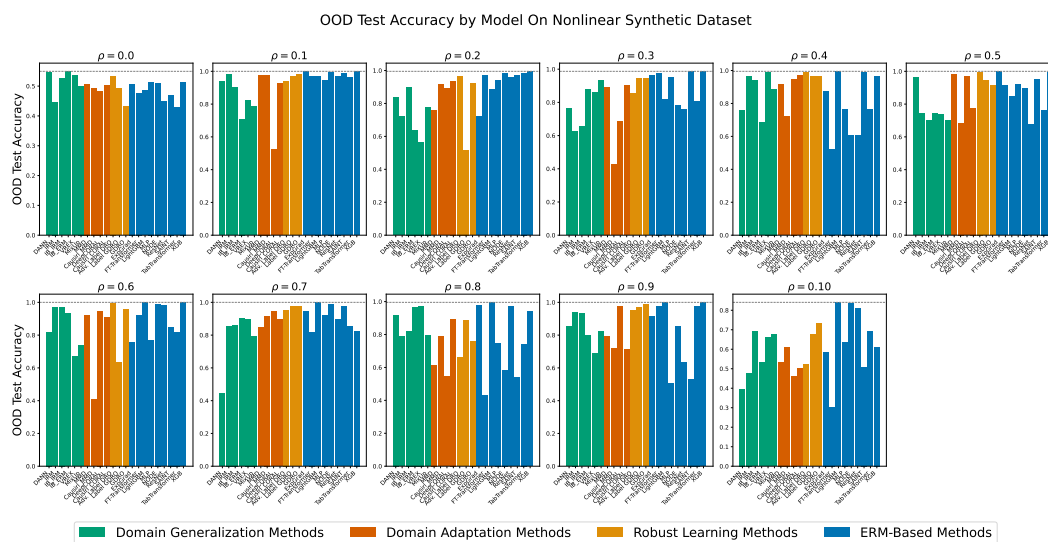


Figure D24: OOD test accuracy in the non-linear setting, reported per model across differing values of ρ

928 E Qualitative analysis of hidden confounding shift in real-world datasets

929 In this section, we undertake a qualitative study of the presence of hidden confounding shifts within
 930 real-world datasets. Tables E3–E18 present the potential hidden confounders that bear upon various
 931 observed covariates across several real-world datasets [57]. For further details concerning these
 932 datasets, we refer the reader to [24, 57]. We enlist GPT4o [1] for this task, acknowledging that the
 933 list of unobserved confounding variables so obtained is by no means exhaustive. This study lends
 934 support to the arguments advanced for the existence and impact of hidden confounding shifts across
 935 domains in real-world datasets.

936 **Prompt:** We employ the prompt below to query GPT4o for possible hidden confounders.

937 For a target variable, you will be given lists of causal, arguably
 938 causal, anti-causal, and spurious covariates. Also, you will be provided
 939 with environment variables such that different datasets are collected in
 940 environments induced by these environment variables. Provide possible
 941 hidden confounders that cause any of the target, causal, arguably causal,
 942 spurious, and anti-causal covariates. That is, the hidden confounders
 943 should influence the distribution of their children in different domains.
 944 Ensure hidden confounders are not a part of the given variables, and your
 945 explanation should include how hidden confounders influence their children
 946 in different environments.

Table E3: **Dataset:** Food stamps. **Target:** Food stamp reciprocity in the past year for households with child. **Environments:** regions in the United States.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Food stamp reciprocity, Age, Sex, Race, Marital status	Poor economic conditions can increase food insecurity (raising food stamp usage), while also influencing population demographics due to migration, employment patterns, and household formation.
State or regional public assistance policies	Food stamp reciprocity, Marital status, Number of children, Household income	Generous or restrictive assistance policies directly affect food stamp eligibility and indirectly influence household structure and income distribution.
Cultural attitudes towards welfare and public assistance	Food stamp reciprocity, Marital status, Disability, Race, Ethnicity	Social stigma or support for welfare affects both the uptake of food stamps and social norms around marriage, disability reporting, and identity categories.
Regional health care conditions	Food stamp reciprocity, Disability, Cognitive difficulty, Hearing difficulty, Vision difficulty	Regions with limited healthcare access may show higher disability prevalence and greater reliance on food stamps due to increased financial and care burdens.

Table E4: **Dataset:** Income. **Target:** Total person’s income $\geq 56K$ for employed adults. **Environments:** regions in the United States.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Income, Occupation, Educational attainment, Marital status	Stronger local economies offer better-paying jobs and education access, increasing income; they also influence occupational choices and household formation patterns.
State or regional tax and labor policies	Income, Class of worker, Educational attainment, Marital status	Tax incentives and labor protections affect wages and employment types, while also shaping decisions about education and family due to financial security.
Cost of living and regional affordability	Income, Occupation, Marital status, Educational attainment	Higher living costs necessitate higher incomes and may drive occupational or educational shifts; they also influence marriage or cohabitation decisions.
Cultural norms and regional economic history	Income, Educational attainment, Occupation	Cultural values and historical industry presence shape education levels and job opportunities, which in turn affect income distributions.

Table E5: **Dataset:** Public coverage. **Target:** Coverage of non-Medicare eligible low-income individuals. **Environments:** Disability statuses.

Hidden confounder	Affected variables	Reason for confounding
State or regional healthcare policies	Public health coverage, Marital status, Employment status, Income	Differences in Medicaid expansion and public insurance eligibility influence health coverage; these policies also affect economic stability, employment, and family dynamics.
Economic conditions and poverty levels	Public health coverage, Employment status, Income, Marital status	Poorer regions tend to have lower employment rates and incomes, which increase reliance on public health coverage and influence marriage and household composition.
Cultural attitudes towards disability and healthcare	Public health coverage, Disability status, Cognitive difficulty, Hearing difficulty, Vision difficulty	Stigma or support for disability and public care varies culturally, affecting both the reporting of disabilities and the likelihood of seeking or receiving public coverage.
Urban vs. rural divide	Public health coverage, Educational attainment, Marital status, Occupation	Urban areas typically offer better healthcare access, education, and jobs, all of which affect coverage likelihood and socioeconomic indicators.

Table E6: **Dataset:** Unemployment. **Target:** Classify whether a person is unemployed. **Environments:** Educational attainments.

Hidden confounder	Affected variables	Reason for confounding
Local economic conditions	Employment status, Occupation, Marital status, Mobility status	Regional economic strength affects job availability and unemployment rates; it also shapes occupation types, migration decisions, and household stability.
State or regional labor market policies	Employment status, Occupation, Educational attainment, Marital status	Labor regulations and unemployment benefits vary by region, affecting hiring practices, education incentives, and family structures.
Cultural and social norms around employment	Employment status, Disability status, Marital status	Cultural attitudes toward work and dependency influence unemployment reporting and societal roles around disability and family responsibilities.
Health and disability status	Employment status, Disability status, Cognitive difficulty, Hearing difficulty, Vision difficulty	Poor health or disabilities reduce the ability to work, directly increasing unemployment and shaping associated health-related variables.

Table E7: **Dataset:** Voting. **Target:** Classify whether a person voted in the U.S. presidential election. **Environments:** United States census regions.

Hidden confounder	Affected variables	Reason for confounding
Political, family, and peer influences	Voted in national election, Party identification, Political participation	Social networks shape political ideology and engagement, influencing both voting likelihood and party alignment.
Media consumption habits	Voted in national election, Party identification, Political knowledge, Voting behavior	Media exposure affects awareness of political issues and biases, influencing party affiliation, political knowledge, and voting participation.
Social capital	Voted in national election, Political participation, Interest in elections	Strong community ties and civic networks increase political interest and participation, leading to higher voter turnout.
Civic education and political engagement programs	Voted in national election, Interest in elections, Political knowledge	Educational programs raise political awareness and civic responsibility, influencing both knowledge levels and voting decisions.
Historical and cultural context	Voted in national election, Party identification, Interest in elections	Historical events and regional political culture affect interest in elections and party alignment, which in turn influence voting behavior.

Table E8: **Dataset:** Hypertension. **Target:** Whether a person has hypertension. **Environments:** Body Mass Index (BMI) values.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status	Income, Employment status, Smoking habits, Alcohol consumption, Physical activity, Healthcare access, Medical cost, High blood pressure diagnosis	Lower socioeconomic status reduces access to healthcare and healthy lifestyle options, leading to poor diet, limited activity, and delayed diagnosis, which jointly influence both BMI and hypertension risk.
Access to healthcare services	Healthcare access, Medical costs, Smoking habits, Alcohol consumption, Physical activity, High blood pressure diagnosis	Limited healthcare access results in underdiagnosis and unmanaged hypertension, while also affecting lifestyle choices that vary with BMI, confounding the relationship between BMI and hypertension.
Genetic predisposition	Age group, Race, Sex, Smoking habits, Diabetes, High blood pressure diagnosis	Genetic risk factors for hypertension may co-vary with demographic attributes and influence both hypertension prevalence and BMI distribution across subpopulations.
Psychosocial stress	Smoking habits, Alcohol consumption, Physical activity, High blood pressure diagnosis, Diabetes, Age group	Chronic stress alters behavior (e.g., smoking, inactivity) and physiological responses, contributing to both increased BMI and elevated blood pressure, thus confounding the BMI–hypertension link.
Environmental factors	Physical activity, Diet, Smoking habits, High blood pressure diagnosis, Diabetes, BMI category	Living environments affect access to recreational spaces, food quality, and pollution exposure, influencing both BMI and hypertension risks. These vary across BMI categories, creating confounding.
Dietary habits beyond fruits and vegetables	Alcohol consumption, Smoking habits, Physical activity, High blood pressure diagnosis, BMI category	High-sodium or processed-food diets raise both BMI and hypertension risk. Variation in such unmeasured dietary habits across BMI categories creates spurious associations with hypertension.

Table E9: **Dataset:** College Scorecard. **Target:** Predict completion rate for first-time, full-time students at four-year institutions. **Environments:** Based on Carnegie Classifications.

Hidden confounder	Affected variables	Reason for confounding
Institutional funding and re-sources	Accreditor, Control of institution, Highest degree awarded, In-state tuition, Out-of-state tuition, Cost of attendance, SAT scores	Wealthier institutions can offer better academic support, facilities, and programs, leading to higher completion rates and more selective admission profiles. This varies across Carnegie classifications.
Regional socio-economic factors	Region, Poverty rate, Unemployment rate, SAT scores	Economic conditions across regions affect affordability, student preparedness, and institutional support levels, all influencing both enrollment outcomes and graduation likelihood.
Demographic factors	HBCU flag, Federal loan recipient rate, Pell grant recipient rate, ACT scores, Undergraduate enrollment	Student demographics shape financial aid needs, academic preparation, and graduation rates. Their effect differs by institution type and selectivity under Carnegie categories.
Community support and engagement	Distance-education flag, Federal loan recipient rate, Pell grant recipient rate, SAT scores, Undergraduate enrollment	Supportive institutional communities improve retention and completion. Variation in engagement across institution types and student aid profiles induces confounding.
Admission selectivity	Admission rate, SAT (reading/math) midpoints, ACT midpoint, Undergraduate enrollment	Selective admissions correlate with better-prepared students and higher completion rates, and vary with institutional prestige and classification.
State and local policies	Region, Poverty rate, Unemployment rate, Cost of attendance	Differences in education funding and public policy affect cost structures and completion outcomes, interacting with institutional classification and regional demographics.

Table E10: **Dataset:** ASSISTments. **Target:** Predict whether a student solves a problem correctly on the first attempt in an online learning tool. **Environments:** Different schools.

Hidden confounder	Affected variables	Reason for confounding
Institutional teaching quality	Hint count, Attempt count, Skill ID, Problem type, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	Variation in instructional quality and pedagogy across schools affects how effectively students engage with content, leading to differences in problem-solving strategies, response behavior, and emotional states.
Student motivation	Hint count, Attempt count, Skill ID, Problem type, First action, Milliseconds to first response, Overlap time, Average confidence	Differences in intrinsic motivation across schools influence students' willingness to persevere, seek help, or give up quickly, affecting interaction and performance.
Classroom environment	Hint count, Attempt count, Tutor mode, Position, Type, First action, Average confidence	Peer dynamics, classroom culture, and noise levels vary by school and affect how confidently and independently students solve problems.
School technology infrastructure	Hint count, Attempt count, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	Access to reliable devices and fast internet differs by school, influencing response time, tool usage, and student experience.
Teacher-student interaction	Hint count, Attempt count, Tutor mode, Position, Type, First action, Milliseconds to first response, Overlap time, Average confidence	The level of teacher guidance and feedback shapes how much support students require during problem-solving, affecting engagement and confidence.
Previous academic performance	Hint count, Attempt count, Skill ID, Problem type, First action, Average confidence	Students' prior achievement affects how easily they solve problems, their need for assistance, and their confidence, all of which vary across schools.

Table E11: **Dataset:** ICU. **Target:** Predict whether the patient will stay in the ICU for longer than 3 days. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, etc.	SES influences access to healthcare, preventive services, and overall health status. Differences in SES across insurance types lead to variability in pre-ICU health, physiological indicators, and ICU stay duration.
Hospital resources and care quality	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Hospital infrastructure and care standards affect monitoring, intervention speed, and clinical decisions. These factors vary by insurance coverage and influence ICU outcomes and vitals.
Comorbidities	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Presence of chronic conditions (e.g., diabetes, cardiovascular disease) affects both the need for prolonged ICU care and physiological measurements. The distribution of comorbidities differs across insurance types.
Insurance-related treatment variability	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, etc.	Differences in treatment timing, intensity, and access to specialists based on insurance policies affect ICU stay duration and clinical metrics.
Genetic factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, etc.	Inherited traits influence predisposition to organ failure, metabolic responses, and recovery trajectories. These effects are partially mediated by ethnicity and age distributions, which vary across insurance groups.

Table E12: **Dataset:** Hospital mortality. **Target:** Classify whether an ICU patient expires in the hospital during their current visit. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Systolic blood pressure, White blood cell count, etc.	SES shapes access to timely and high-quality care, preventive services, and general health status. Patients with higher SES often have better insurance and outcomes, leading to confounding with mortality risk.
Comorbidities	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Systolic blood pressure, etc.	Pre-existing conditions such as diabetes or heart disease increase mortality risk and influence physiological features. Their prevalence differs by insurance type, creating confounding.
Hospital resources and care quality	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Systolic blood pressure, White blood cell count, etc.	Access to advanced treatments, trained staff, and timely interventions influences survival rates. These factors correlate with insurance coverage, confounding mortality outcomes.
Genetic factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, etc.	Genetic predispositions affect disease susceptibility and treatment responses. Variations in genetic risk factors may correlate with demographic traits across insurance types.
Lifestyle and behavioral factors	Age, Gender, Ethnicity, Height, Weight, Bicarbonate, CO ₂ , pCO ₂ , pO ₂ , Lactate, Sodium, Hemoglobin, Oxygen saturation, Respiratory rate, Heart rate, Temperature, Systolic blood pressure, White blood cell count, etc.	Behaviors such as smoking, diet, and physical activity affect long-term health and mortality risk. These behaviors vary systematically with SES and insurance coverage, influencing both target and physiological features.

Table E13: **Dataset:** Childhood lead. **Target:** Predict blood lead levels above CDC blood level reference value. **Environments:** Poverty-income ratios.

Hidden confounder	Affected variables	Reason for confounding
Environmental Exposure	Blood lead levels, Country of birth, Age, Race and Hispanic origin	Environmental exposure to lead influences blood lead levels, and this varies significantly across socio-economic groups. People in lower PIR groups are more likely to live in areas with higher lead contamination, which contributes to higher blood lead levels. Furthermore, environmental factors may affect the demographic distribution (e.g., country of birth, race).
Access to Healthcare	Blood lead levels, Age, Gender, Race and Hispanic origin, Marital status, Education	Limited access to healthcare, especially in lower PIR groups, means fewer opportunities for detection and treatment of lead poisoning. This results in higher blood lead levels, with disparities also influencing demographic variables like age, gender, and education. Additionally, healthcare access varies by insurance and socio-economic status, further confounding the relationships.
Diet and Nutrition	Blood lead levels, Age, Gender, Race and Hispanic origin	Dietary factors, such as poor nutrition in lower PIR groups, can exacerbate lead absorption. Malnutrition increases the body's susceptibility to lead poisoning, raising blood lead levels. In contrast, higher PIR groups may have better access to nutritious foods, lowering lead absorption, thus creating a confounding effect in how socio-economic status and race influence lead toxicity.
Housing Conditions	Blood lead levels, Country of birth, Race and Hispanic origin, Marital status	Older housing conditions, which are more prevalent in lower PIR groups, contribute significantly to elevated lead exposure (e.g., lead paint, poor plumbing). These living conditions directly influence blood lead levels and can also correlate with demographic factors like country of birth, race, and marital status. This introduces confounding, as socio-economic status impacts both exposure and the demographics of affected individuals.
Occupation	Blood lead levels, Age, Race and Hispanic origin, Marital status, Education	Certain occupations, which are more common among lower PIR groups, involve higher lead exposure (e.g., construction, manufacturing). Occupational lead exposure directly impacts blood lead levels and is often correlated with education, marital status, and socio-economic status. The varying prevalence of lead exposure by occupation introduces confounding, especially across different PIR groups.

Table E14: **Dataset:** Diabetes. **Target:** Predict diabetes. **Environments:** Preferred race categories.

Hidden confounder	Affected variables	Reason for confounding
Genetic predisposition	Diabetes, BMI, High blood pressure, High blood cholesterol	Genetic factors and family history contribute to both the onset of diabetes and comorbid conditions like obesity, hypertension, and high cholesterol. These genetic predispositions can make individuals more susceptible to diabetes, leading to confounding as they correlate with other health indicators.
Access to healthcare	Diabetes, Physical health, BMI, Healthcare coverage, Health checkups	Limited or unequal access to healthcare, especially in marginalized racial groups, leads to disparities in diabetes diagnosis, management, and comorbidity treatment. It also influences the frequency of health checkups and access to medications, which can confound the relationship between diabetes status and other health metrics.
Dietary habits and food availability	Diabetes, BMI, Physical health, Alcohol consumption, Fruit and vegetable intake	Dietary habits, often shaped by socioeconomic status and local food environments, influence weight, health behaviors (such as alcohol consumption), and diabetes risk. People in lower socioeconomic strata may have limited access to healthy food options, leading to higher BMI and increased diabetes risk, creating confounding effects on health outcomes.
Psychosocial stress and mental health factors	Diabetes, Mental health, Physical health, BMI, Physical activity, Doctor visits	Chronic stress and mental health issues, often higher in marginalized groups, contribute to diabetes development and complicate its management. These factors also affect physical health (e.g., weight gain due to stress) and health-seeking behaviors (e.g., fewer doctor visits), leading to confounding by influencing both diabetes risk and its associated variables.
Socioeconomic status beyond income	Diabetes, Income, Physical health, Healthcare coverage, Education level	Socioeconomic factors, such as occupation, education, and neighborhood wealth, influence access to healthcare, nutrition, and overall health behaviors. These factors can confound the relationships between diabetes and other socio-economic variables like income and education, as they shape opportunities for prevention and treatment.

Table E15: **Dataset:** Sepsis. **Target:** Predict, from a set of fine-grained ICU data, whether a patient will experience sepsis onset within the next 6 hours. **Environments:** Lengths of ICU stay.

Hidden confounder	Affected variables	Reason for confounding
Infection prevalence in ICU	SepsisLabel, Temperature (Temp), Leukocyte count (WBC), Heart rate (HR), Blood urea nitrogen (BUN)	Higher infection rates in certain ICU units can lead to a higher probability of sepsis onset (SepsisLabel). These infection rates influence biomarkers such as WBC, HR, and BUN, creating confounding because the unit's infection environment affects both the likelihood of sepsis and the observed clinical measures.
Quality of ICU Care	SepsisLabel, Fibrinogen concentration (Fibrinogen), Leukocyte count (WBC), Platelet count (Platelets)	Higher-quality care in certain ICUs may lead to earlier identification and treatment of sepsis, resulting in more accurate SepsisLabel predictions. Additionally, better care could affect biomarkers like fibrinogen, WBC, and platelets, which are critical in sepsis detection and progression, thereby confounding the relationships between these variables and the outcome.
Severity of underlying conditions	SepsisLabel, Blood urea nitrogen (BUN), Creatinine, Lactate, Calcium	Patients with severe chronic conditions (e.g., kidney disease, cardiovascular issues) are at a higher risk of sepsis and may show abnormal levels in biomarkers like BUN, creatinine, lactate, and calcium. These underlying conditions contribute to the SepsisLabel outcome and confound the relationship between the biomarkers and the likelihood of sepsis, varying across ICU units depending on patient population.
Patient's socio-economic status	Age (Age), Gender (Gender), Leukocyte count (WBC), Fibrinogen concentration (Fibrinogen), Platelet count (Platelets)	Socio-economic factors, such as access to healthcare, can influence both the likelihood of sepsis and the observed clinical biomarkers. For example, patients from lower socio-economic backgrounds may have delayed hospitalizations or inadequate care, which affects both SepsisLabel and the progression of sepsis as indicated by WBC, fibrinogen, and platelet levels.
Hospital-specific protocols and treatment guidelines	SepsisLabel, Lactate, Glucose, Creatinine	Differences in hospital protocols for sepsis treatment, such as timing of interventions and choice of sepsis bundles, can affect both the SepsisLabel and biomarkers like lactate, glucose, and creatinine. These protocols lead to variability in how sepsis is diagnosed and treated across different hospitals and ICU units, confounding the relationship between biomarkers and sepsis outcomes.

Table E16: **Dataset:** Hospital readmission. **Target:** Predict whether a diabetic patient is readmitted to the hospital within 30 days of their initial release. **Environments:** Admission sources.

Hidden confounder	Affected variables	Reason for confounding
Socio-economic status (SES)	Race, Gender, Age, Payer code, Medical specialty, Number of outpatient visits, Number of emergency visits, Number of inpatient visits, Diabetes medication prescribed	SES influences access to healthcare, patient demographics, and chronic disease rates. It can also determine the type of care received based on admission source (e.g., emergency department vs. outpatient settings). Differences in healthcare access, such as availability of medications, may impact readmission rates and associated variables like outpatient visits and prescribed medication.
Severity of illness	Primary diagnosis, Secondary diagnosis, Number of diagnoses, Discharge type, Medication changes (e.g., Insulin, Glipizide)	More severe illness increases the likelihood of readmission and influences the complexity of diagnoses and the treatments administered. The severity of illness may differ based on the admission source (e.g., emergency versus outpatient), impacting the number and type of diagnoses and treatments prescribed at discharge, affecting readmission likelihood.
Access to healthcare resources	Time in hospital, Discharge disposition, Number of procedures, Number of medications, Number of lab tests	Access to healthcare resources (e.g., time in hospital, availability of procedures and medications) influences treatment decisions and outcomes. Different admission sources may have varying levels of available resources, leading to different lengths of stay, the types of procedures performed, and overall treatment quality, which can affect the likelihood of readmission.
Patient's adherence to medication	Change in medications, Diabetes medication prescribed, Number of outpatient visits, Number of emergency visits	Adherence to prescribed medications is often influenced by socio-economic status, which can vary depending on admission source. Non-adherence may lead to medication changes, affecting diabetes control and subsequent readmission risk. The number of outpatient and emergency visits can also reflect how well a patient manages their diabetes and the likelihood of complications.
Hospital-specific protocols	Time in hospital, Number of diagnoses, Discharge disposition, Readmitted	Different hospitals and healthcare systems implement various protocols for discharge planning and readmission prevention, which can affect readmission rates. These protocols may vary by admission source, where patients admitted via the emergency department may receive different follow-up instructions and care than those admitted through other channels.

Table E17: **Dataset:** MEPS. **Target:** Measure of health care utilization. **Environments:** Insurance types.

Hidden confounder	Affected variables	Reason for confounding
Socioeconomic status (SES)	Years of education, Employment status, Hourly wage, Paid sick leave, Paid leave to visit doctor, Family size, Insurance coverage, Healthcare utilization	SES can influence access to healthcare services, insurance coverage, and the ability to use medical services. Insurance types often correlate with SES levels, where individuals with lower SES may be more likely to be insured by government programs (e.g., Medicaid), which in turn impacts healthcare utilization patterns across different SES groups.
Healthcare access	Paid sick leave, Insurance coverage, Employer offers health insurance, Healthcare utilization	Limited access to healthcare, such as lack of insurance or paid sick leave, directly impacts healthcare utilization. The type of insurance a person holds is often tied to access to various medical services. The level of coverage and accessibility differs across insurance types, influencing healthcare behaviors such as whether a patient can afford and utilize healthcare services.
Health behaviors	Perceived health status, Asthma medications, Limitations in physical functioning, Healthcare utilization	Lifestyle factors like smoking, alcohol consumption, and physical activity can directly affect health status and healthcare needs. Health behaviors differ across groups with different insurance types, and these behaviors contribute to healthcare utilization. Insurance coverage can also be influenced by perceived health status, which varies across insured groups, affecting utilization of medical services.
Chronic health conditions	Asthma medications, Perceived health status, Limitations in physical functioning, Healthcare utilization	Chronic conditions like asthma or diabetes increase the need for healthcare services, leading to higher healthcare utilization. People with chronic conditions are more likely to be covered by Medicare or Medicaid, which influences the type of insurance they hold and impacts their healthcare utilization and associated features (e.g., medications and physical limitations).
Regional healthcare infrastructure	Region, Family size, Healthcare utilization, Paid sick leave, Paid leave to visit doctor, Employment status	The quality and availability of healthcare infrastructure vary across regions, which can impact healthcare utilization. In underserved regions, people may be more reliant on public insurance options like Medicaid, affecting healthcare access and behaviors. The regional differences in healthcare systems can lead to disparities in access to medical services, insurance coverage, and utilization of healthcare.

Table E18: **Dataset:** Poverty. **Target:** Predict household income-to-poverty ratio. **Environments:** Citizenship statuses.

Hidden confounder	Affected variables	Reason for confounding
Social capital	Income, Unemployment compensation, Disability benefits, Family size, Housing conditions, Household income-to-poverty ratio	Stronger social support networks can improve access to income, benefits, and resources like unemployment or disability compensation. Social capital can vary based on citizenship status, which in turn influences income, housing conditions, and eligibility for benefits, confounding the relationship between household income and poverty ratios.
Workplace discrimination or bias	Household income, Worker's compensation, Unemployment compensation, Pension, Family income, Disability benefits	Discrimination in the workplace can limit opportunities for higher wages, pensions, or disability benefits, especially for marginalized groups (e.g., racial minorities, non-citizens). This bias varies by citizenship status and leads to biased associations between income, benefits, and poverty levels.
Healthcare access	Household income, Health insurance premiums, Medical expenses, Disability benefits, Family size, Health conditions, Medicaid/Medicare assistance	Limited access to healthcare, particularly for non-citizens, can lead to higher medical expenses and greater reliance on public health assistance. Citizenship status directly impacts eligibility for public healthcare programs, confounding relationships between medical expenses and income, and affecting poverty ratios.
Access to education and skills development	Educational attainment, Household income, Employment status, Unemployment compensation, Income from assistance, Savings	Disparities in educational opportunities, often linked to citizenship status, influence income potential and eligibility for government assistance. These disparities, in turn, affect the household income-to-poverty ratio and employment outcomes, creating confounding relationships between education and income.
Cultural factors	Family size, Household income, Savings, Disability benefits, Healthcare utilization, Income assistance, Living arrangements	Cultural factors influence financial management, family support, and the use of social programs. These factors can vary across citizenship statuses, leading to differences in how household income and assistance are distributed, and thus confounding the relationship between income, benefits, and financial behavior.
Housing market and rent conditions	Housing ownership, Housing costs, Family size, Household income, Rent payments, Social security benefits, Medical aid	Housing market conditions, especially rent disparities, can create financial strain and impact the household income-to-poverty ratio. Citizenship status often influences eligibility for housing assistance and rent subsidies, which confounds the relationship between housing costs and income, particularly in areas with significant immigrant populations.